LAB 0: VirtualBox Network Mode Comparison - Kali & Metasploitable2

Comparison of NAT, NAT Network, Bridged, Host-only, and Internal Network modes

Wednesday, July 15, 2026

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1. Objective:

This lab investigates how VirtualBox's five network adapter modes i.e. NAT, NAT Network, Bridged, Host-Only, and Internal Network; affect connectivity between two virtual machines, Kali Linux and Metasploitable2, along with their reachability from the host machine and access to the internet. Each mode was configured and tested individually using live connectivity checks (ip a, ping), with results captured as screenshots rather than assumed documentation. The objective is to establish a correctly understood, deliberately isolated lab network as the foundation for later offensive and defensive security labs, and to build a practical understanding of network isolation, since misunderstanding network boundaries is a common source of both misconfigured lab environments and real detection blind spots.

2. Lab Environment:

The Environment used is as follows:

2.1: Host Machine:

- OS: Windows 11
- CPU: 13th Gen Intel Core i7-1355U, 1.70 GHz
- RAM: 8.00 GB

2.2: Hypervisor:

- Oracle VirtualBox 7.2.12

2.3: Kali Linux:

- Version: kali-linux-2026.2-virtualbox-amd64
- RAM Allocated: 2048 MB
- CPUs Allocated: 2

2.4: Metasploitable2:

- Version: Metasploitable2 (Ubuntu 8.04-based)
- RAM Allocated: 512 MB
- CPUs Allocated: 1

Kali is the attacker and testing machine and metasploitable2 is intentionally vulnerable target machine, these roles will be used in subsequent labs

3. Networking Concepts Primer:

This section explains the key networking concepts referenced throughout the lab, to ensure the results are understandable without prior context:

3.1: Reading 'ip a' Output:

The 'ip a' command lists every network interface on the machine along with its current configuration. Below is annotated output taken from the Host-only Adapter test (Metaspoitable2):

```
2: eth0: <BROADCAST ,MULTICAST ,UP ,LOWER_UP> mtu 1500 qdisc
 pfifo_fast qlen 1000
      link/ether 08:00:27:59:a3:88 brd ff:ff:ff:ff:ff:ff
      inet 192.168.56.101/24 brd 192.168.56.255 scope global
 eth0
```

- **`2:`** an internal index number the OS assigns to this interface.
- **`eth0:`** the interface name (explained in 3.2)
- **`<BROADCAST ,MULTICAST ,UP ,LOWER_UP>:`** flags describing the interface's states and capabilities:
  - **`BROADCAST:`** capable of sending broadcast traffic e.g. ARP requests.
  - **`MULTICAST:`** capable of sending multicast traffic
  - **`UP:`** the interface has been administratively enabled
  - **`LOWER_UP:`** the underlying link is actually active and connected
- **`mtu 1500:`** Maximum Transmission Unit; the largest packet size (in bytes) the interface will send without fragmenting it.
- **`link/ether:`** indicates the following value is a MAC address.
- **`inet:`** signals the line contains IPv4 addressing information.

3.2: Interfaces, MAC Addresses, and eth0:

'**eth0**' is Linux's name for the machine's first Ethernet network interface: the "door" through which network traffic enters and leaves. If a second interface existed it would typically be named '**eth1**', and so on. Other common interface names include '**lo**'(loopback) and '**wlan0**' (Wi-Fi).

Each interface also has **a MAC address**; a hardware level address. Shown after **link/ether**. This is the exact address ARP resolves an IP address to. The brd **ff:ff:ff:ff:ff:ff** value is the broadcast MAC address; sending to this address reaches every device on local network segment.

3.3: Subnet Masks and CIDR Notation:

An IP address has two logical parts: a **network portion** and a **host portion**. The subnet mask defines where that split occurs.

192.168.56.101/24 means the **first 24 bits** (the first three octets: 192.168.56) identify the network and the remaining **8 bits** (the last octet) identify individual host within it, giving 256 possible addresses (.0 to .255) with **.0** and **.255** conventionally reserved.

This is why two machines must share the same subnet to communicate directly:**192.168.56.101** and **192.168.56.102** share the same first three octets, so they are recognized as being on same local network. A machine with an address like **192.168.200.10** would be on different network entirely and require a router to reach.

netmask 255.255.255.0 (used in the manual ifconfig commands during the Internal Network test) is the longhand equivalent of /24: each **255 marks a fully "network"** octet, **and 0 marks a fully "host"** octet.

4. Methodology

To ensure consistency and comparability across all five network modes, the following procedure was applied identically to each mode tested:

1. **Power off both VMs.** VirtualBox requires both machines to be fully powered off before the network adapter mode can be changed.
2. **Configure Adapter 1** on both Kali Linux and Metasploitable2 to the mode under test.
3. **Boot both VMs** and run ip a on each to record whether an IP address was assigned automatically, and what address range was used.
4. **Test VM-to-VM connectivity** in both directions using ping -c 4 <target-ip>, once valid IP addresses were confirmed on both machines.
5. **Test internet access** from each VM independently using ping -c 4 8.8.8.8.
6. **Test host-to-VM reachability** by pinging both VMs from the Windows host machine, outside of any VM.
7. **Capture screenshots** of every command and its output as evidence for each step.
8. **Record results** in a comparison table, followed by a written analysis explaining the underlying networking mechanism responsible for each observed result — not simply whether a test passed or failed.

Minor setup steps specific to individual modes (e.g. creating a NAT Network before use, selecting a physical host adapter for Bridged mode, or manually assigning static IP addresses for Internal Network due to the absence of a DHCP server) are documented within that mode's respective section rather than repeated here.

5. Mode 1: NAT

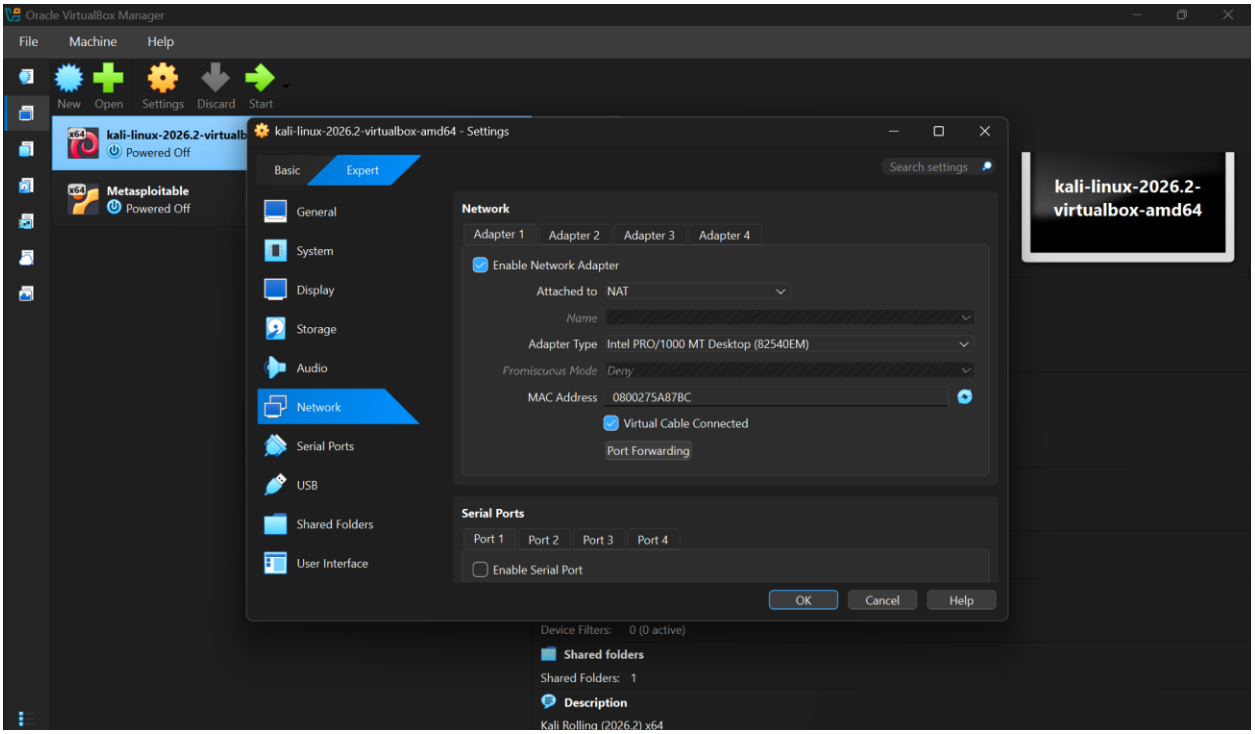
5.1: Configuration:

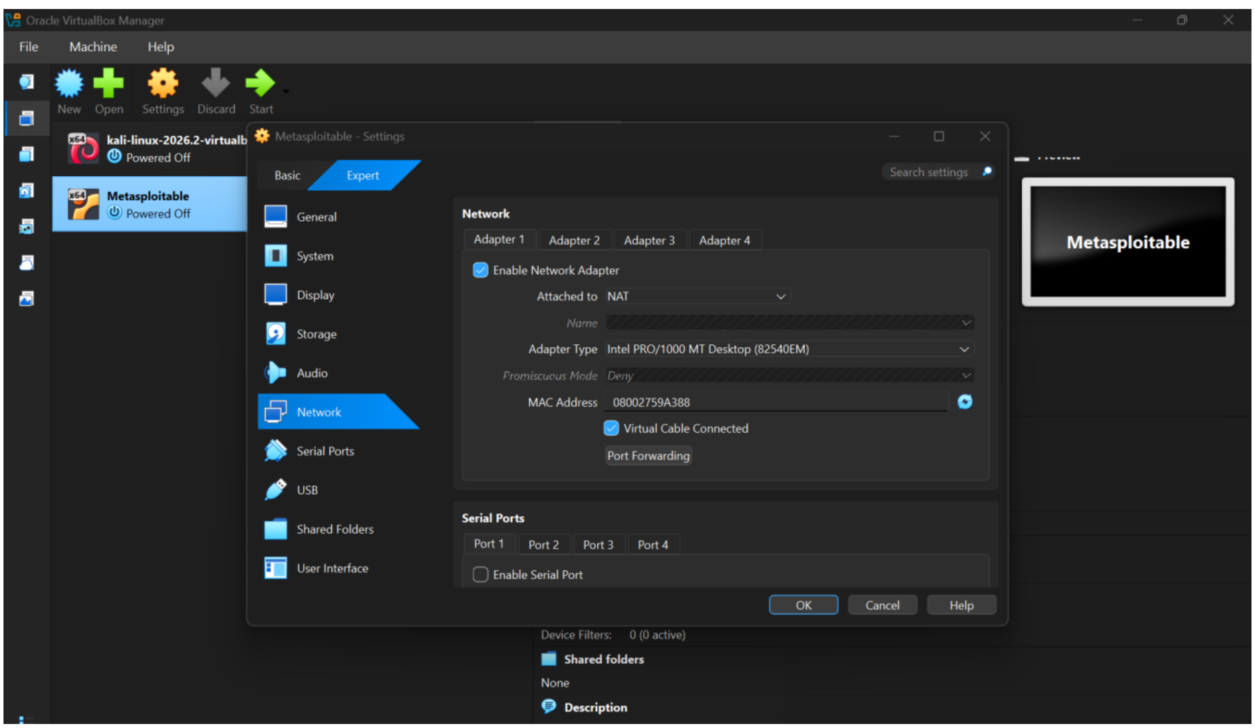
Both Kali and Metasploitable2 set to NAT (VirtualBox default)

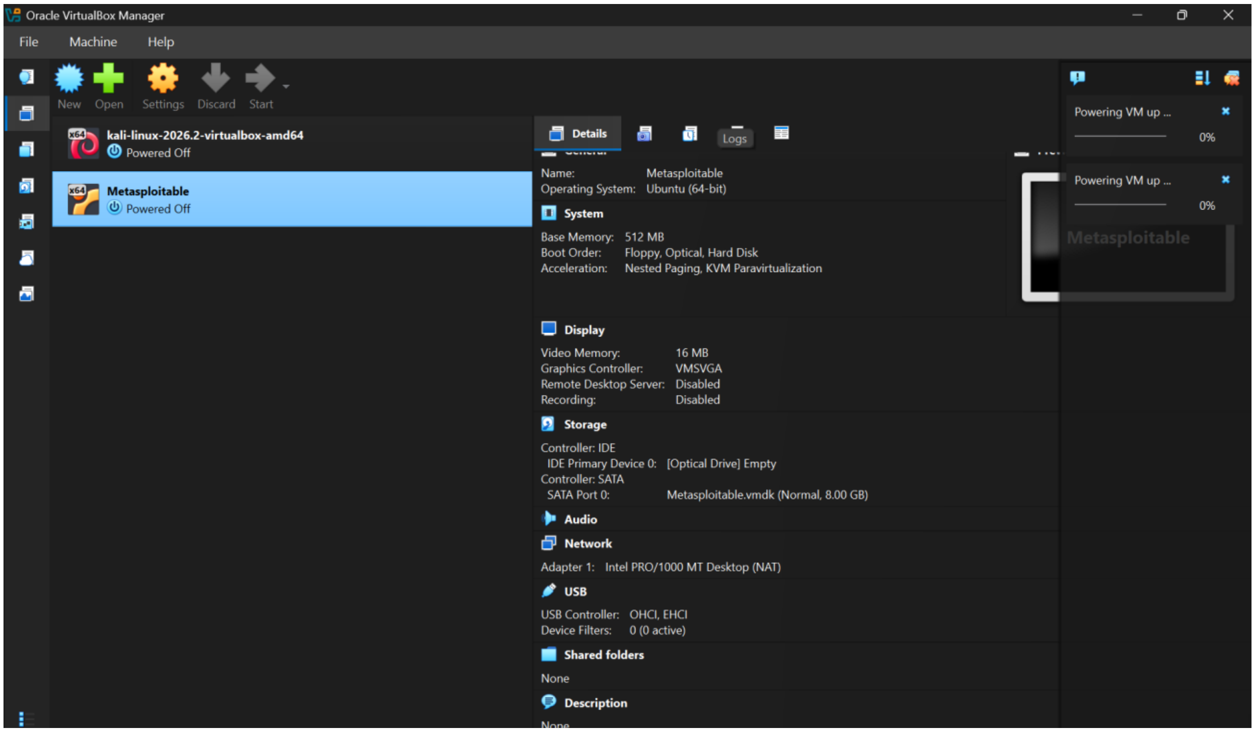
5.2: Architecture Diagram:

```
Internet
  |
Host (VirtualBox NAT Engine)
  |
  ├── Kali (10.0.2.15)  → [isolated bubble]
  |
  └── Metasploitable2 (10.0.2.15) → [separate isolated bubble]

  No path between Kali and Metasploitable2
```

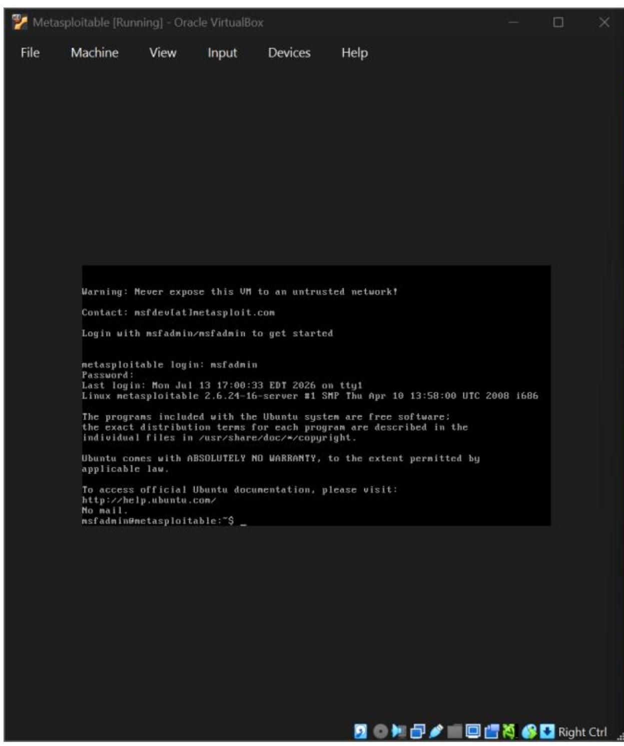
5.3: Screenshots:





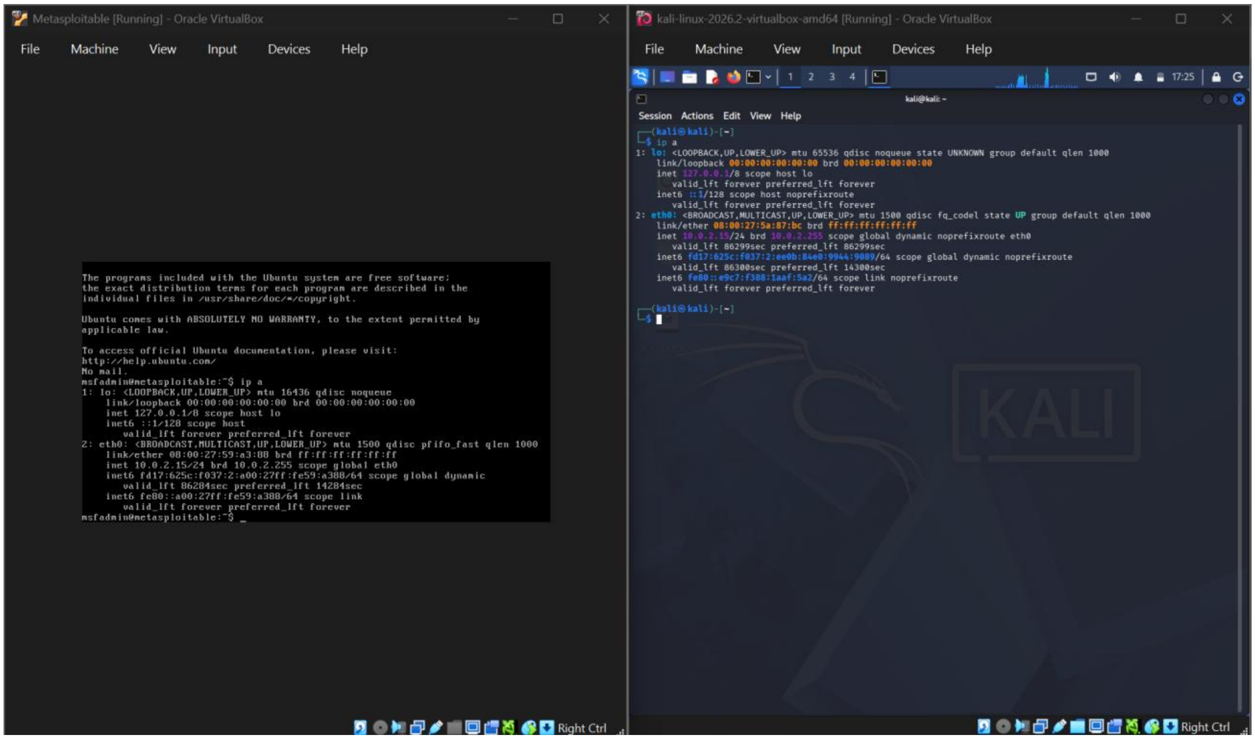


Oracle VirtualBox Manager
File
Machine
Help
New
Open
Settings
Discard
Start
kali-linux-2026.2-virtualbox-amd64
Powered Off
Metasploitable
Powered Off
Details
Logs
Powering VM up ...
0%
Powering VM up ...
0%
Metasploitable
Name: Metasploitable
Operating System: Ubuntu (64-bit)
System
Base Memory: 512 MB
Boot Order: Floppy, Optical, Hard Disk
Acceleration: Nested Paging, KVM Paravirtualization
Display
Video Memory: 16 MB
Graphics Controller: VMSVGA
Remote Desktop Server: Disabled
Recording: Disabled
Storage
Controller: IDE
IDE Primary Device 0: [Optical Drive] Empty
Controller: SATA
SATA Port 0: Metasploitable.vmdk (Normal, 8.00 GB)
Audio
Network
Adapter 1: Intel PRO/1000 MT Desktop (NAT)
USB
USB Controller: OHCI, EHCI
Device Filters: 0 (0 active)
Shared folders
None
Description
None

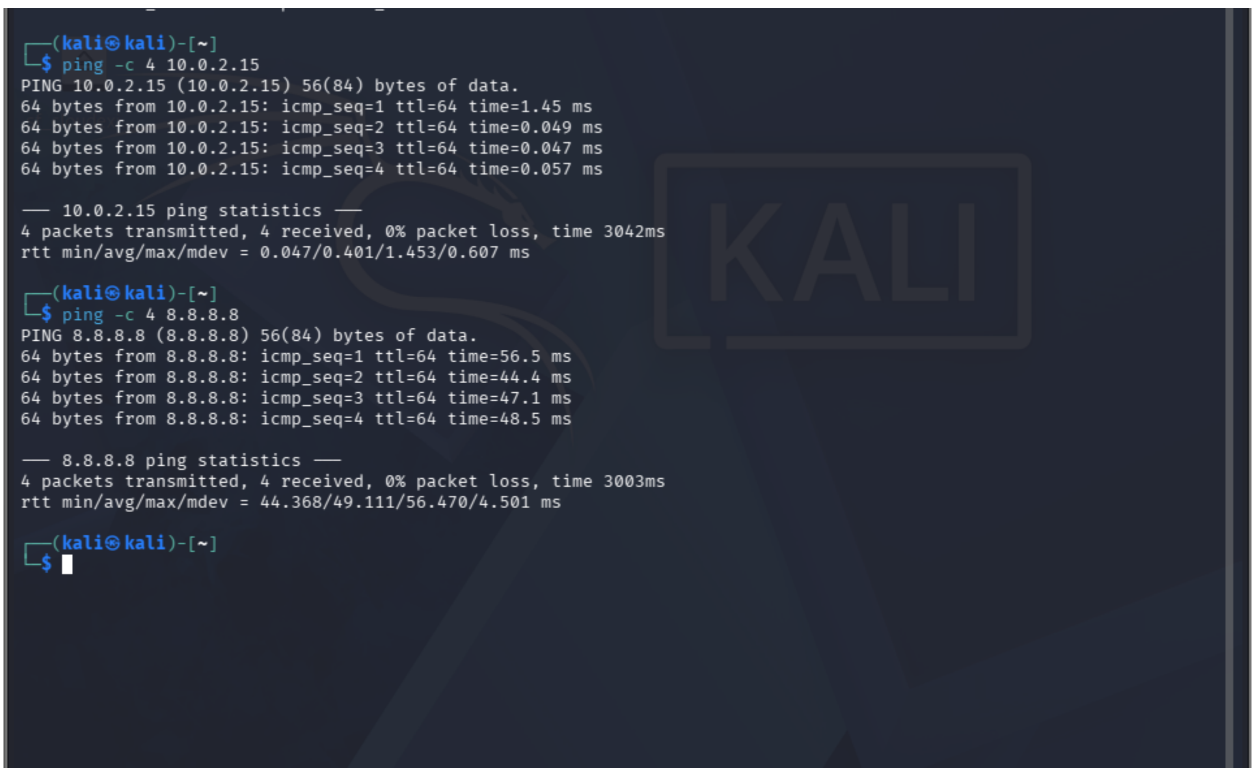


Metasploitable [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Warning: Never expose this VM to an untrusted network!
Contact: msfdev[at]metasploit.com
Login with msfadmin/msfadmin to get started
metasploitable login: msfadmin
Password:
Last login: Mon Jul 13 17:00:33 EDT 2026 on tty1
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ _
Right Ctrl

kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
1 2 3 4
17:25
Home
File System
Trash
sf_cloudexl...
KALI
Right Ctrl



Metasploitable [Running] - Oracle VirtualBox
File Machine View Input Devices Help
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
inet6 ::1/128 scope host
valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
link/ether 08:00:27:59:a3:88 brd ff:ff:ff:ff:ff:ff
inet 10.0.2.15/24 brd 10.0.2.255 scope global eth0
inet6 fd17:625c:f037:2:a00:27ff:fe59:a388/64 scope global dynamic
valid_lft 86284sec preferred_lft 14284sec
inet6 fe80::a00:27ff:fe59:a388/64 scope link
valid_lft forever preferred_lft forever
msfadmin@metasploitable:~$ _
Right Ctrl
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali: ~
Session Actions Edit View Help
(kali@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff
inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
valid_lft 86299sec preferred_lft 86299sec
inet6 fd17:625c:f037:2:ee0b:84e0:9944:9089/64 scope global dynamic noprefixroute
valid_lft 86300sec preferred_lft 14300sec
inet6 fe80::e9c7:f388:1aaf:5a2/64 scope link noprefixroute
valid_lft forever preferred_lft forever
(kali@kali)-[~]
$
KALI
Right Ctrl



(kali㉿kali)-[~]
$ ping -c 4 10.0.2.15
PING 10.0.2.15 (10.0.2.15) 56(84) bytes of data.
64 bytes from 10.0.2.15: icmp_seq=1 ttl=64 time=1.45 ms
64 bytes from 10.0.2.15: icmp_seq=2 ttl=64 time=0.049 ms
64 bytes from 10.0.2.15: icmp_seq=3 ttl=64 time=0.047 ms
64 bytes from 10.0.2.15: icmp_seq=4 ttl=64 time=0.057 ms
— 10.0.2.15 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3042ms
rtt min/avg/max/mdev = 0.047/0.401/1.453/0.607 ms
(kali㉿kali)-[~]
$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=64 time=56.5 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=64 time=44.4 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=64 time=47.1 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=64 time=48.5 ms
— 8.8.8.8 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 44.368/49.111/56.470/4.501 ms
(kali㉿kali)-[~]
$
KALI

```
No mail.
msfadmin@metasploitable:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 08:00:27:59:a3:88 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global eth0
    inet6 fd17:625c:f037:2:a00:27ff:fe59:a388/64 scope global dynamic
       valid_lft 86284sec preferred_lft 14284sec
    inet6 fe80::a00:27ff:fe59:a388/64 scope link
       valid_lft forever preferred_lft forever
msfadmin@metasploitable:~$ ping -c 4 10.0.2.15
PING 10.0.2.15 (10.0.2.15) 56(84) bytes of data.
64 bytes from 10.0.2.15: icmp_seq=1 ttl=64 time=0.116 ms
64 bytes from 10.0.2.15: icmp_seq=2 ttl=64 time=0.058 ms
64 bytes from 10.0.2.15: icmp_seq=3 ttl=64 time=0.055 ms
64 bytes from 10.0.2.15: icmp_seq=4 ttl=64 time=0.118 ms

--- 10.0.2.15 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 2998ms
rtt min/avg/max/mdev = 0.055/0.086/0.118/0.032 ms
msfadmin@metasploitable:~$
```

```
    inet6 fd17:625c:f037:2:a00:27ff:fe59:a388/64 scope global dynamic
       valid_lft 86284sec preferred_lft 14284sec
    inet6 fe80::a00:27ff:fe59:a388/64 scope link
       valid_lft forever preferred_lft forever
msfadmin@metasploitable:~$ ping -c 4 10.0.2.15
PING 10.0.2.15 (10.0.2.15) 56(84) bytes of data.
64 bytes from 10.0.2.15: icmp_seq=1 ttl=64 time=0.116 ms
64 bytes from 10.0.2.15: icmp_seq=2 ttl=64 time=0.058 ms
64 bytes from 10.0.2.15: icmp_seq=3 ttl=64 time=0.055 ms
64 bytes from 10.0.2.15: icmp_seq=4 ttl=64 time=0.118 ms

--- 10.0.2.15 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 2998ms
rtt min/avg/max/mdev = 0.055/0.086/0.118/0.032 ms
msfadmin@metasploitable:~$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=64 time=52.3 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=64 time=43.6 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=64 time=44.6 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=64 time=45.3 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 2999ms
rtt min/avg/max/mdev = 43.660/46.483/52.336/3.429 ms
msfadmin@metasploitable:~$ _
```

5.4: Result Table:

Test	Result
Kali IP	10.0.2.15/24
Metasploitable2 IP	10.0.2.15/24 (identical)
Kali → 10.0.2.15	Self-response 0.047ms avg
Metasploitable2 → 10.0.2.15	Self-response 0.086ms avg
Kali → 8.8.8.8	Success, 0% loss
Metasploitable2 → 8.8.8.8	Success, 0% loss
Host → VM	Not tested (Not applicable to NAT)

5.5: Analysis

Under NAT mode, VirtualBox creates a separate private virtual network for each virtual machine (VM) rather than placing multiple VMs on a shared network. As a result, both Kali Linux and Metasploitable2 were assigned the default IP address 10.0.2.15 without causing an address conflict, because each VM exists within its own isolated virtual network. Although the IP addresses appear identical, they belong to different network instances and are therefore independent.

When Kali attempted to ping 10.0.2.15, the request was resolved within Kali's own private network. Since no other devices were present on that network, Kali effectively pinged itself and received a successful reply, evidenced by the sub-millisecond response time, since the packets never left the local machine and did not traverse an actual network path.

Despite this isolation, both VMs were able to access the internet successfully. VirtualBox's NAT engine translates each VM's outbound traffic and routes it through the host system's network connection, allowing internet connectivity while preventing direct communication between VMs.

5.6: Conclusion:

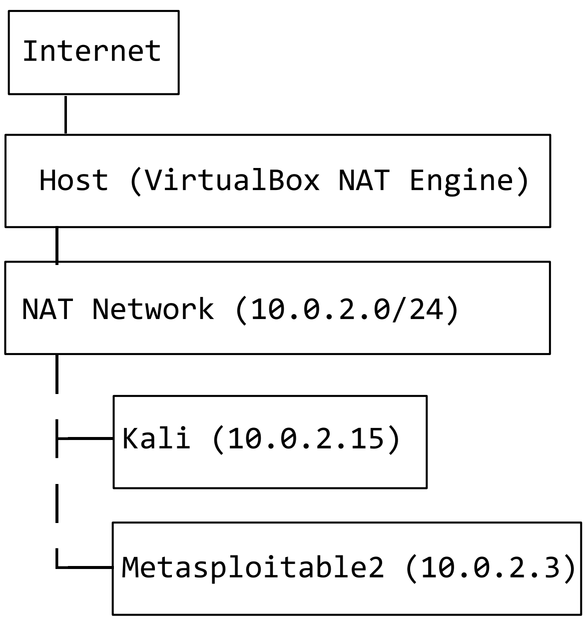
NAT mode is not suitable for lab environments that require direct communication between virtual machines. This limitation is inherent to the design of NAT mode rather than a configurable setting. Because each VM is placed on its own isolated private network, VMs cannot discover or communicate with one another, even if they are assigned the same IP address. While NAT mode provides secure, independent internet access for each VM, it does not support VM-to-VM networking, making it unsuitable for any lab setup requiring communication between multiple machines.

6. Mode 2: NAT Network

6.1: Configuration:

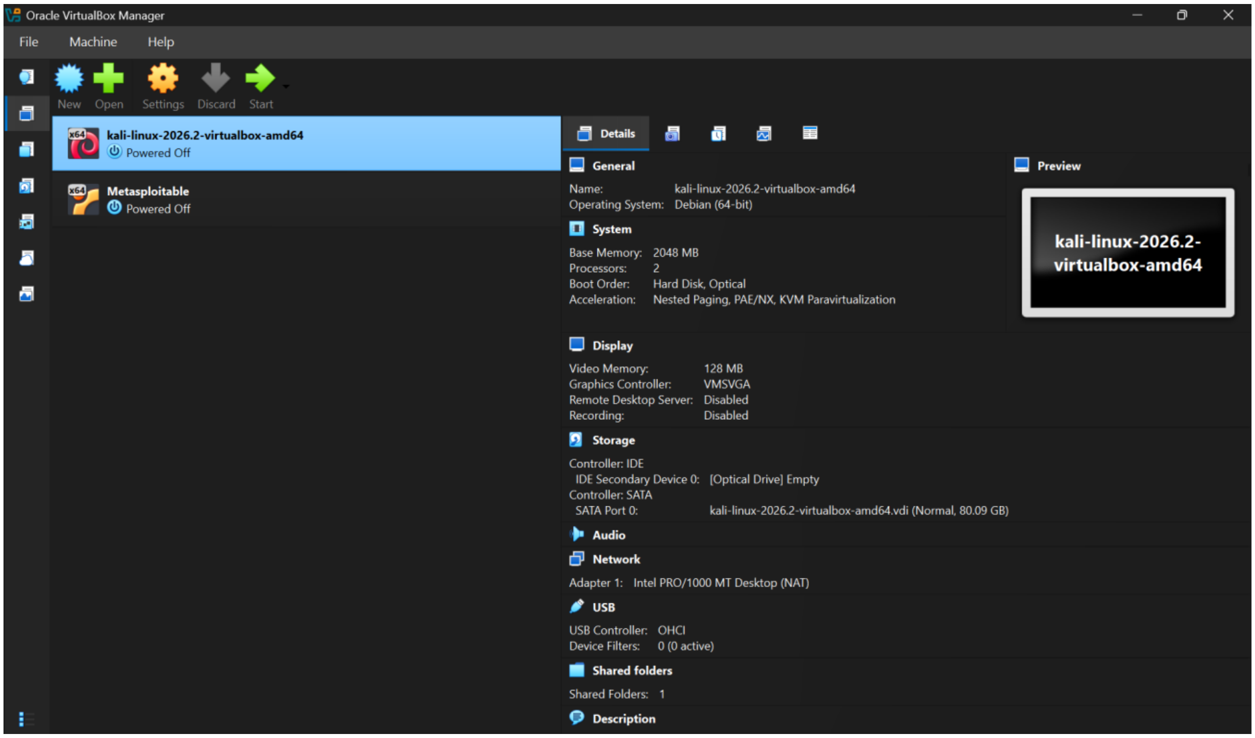
Both Kali and Metasploitable2 set to NAT Network

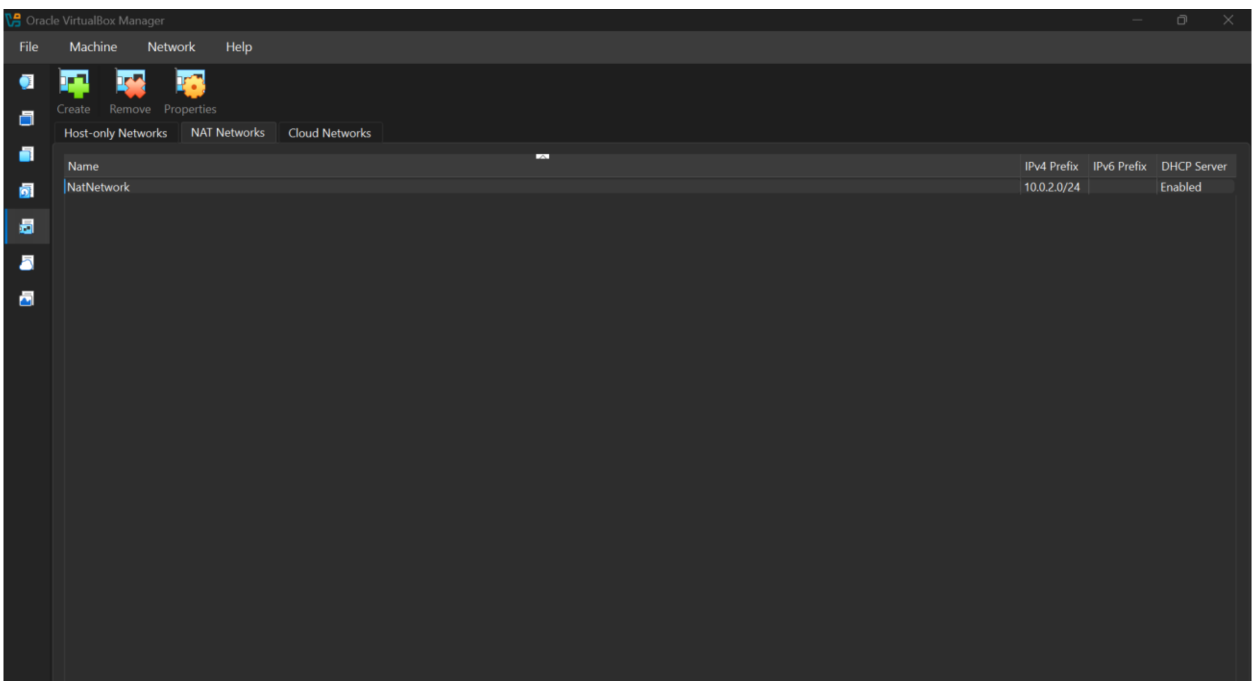
6.2: Architecture Diagram:

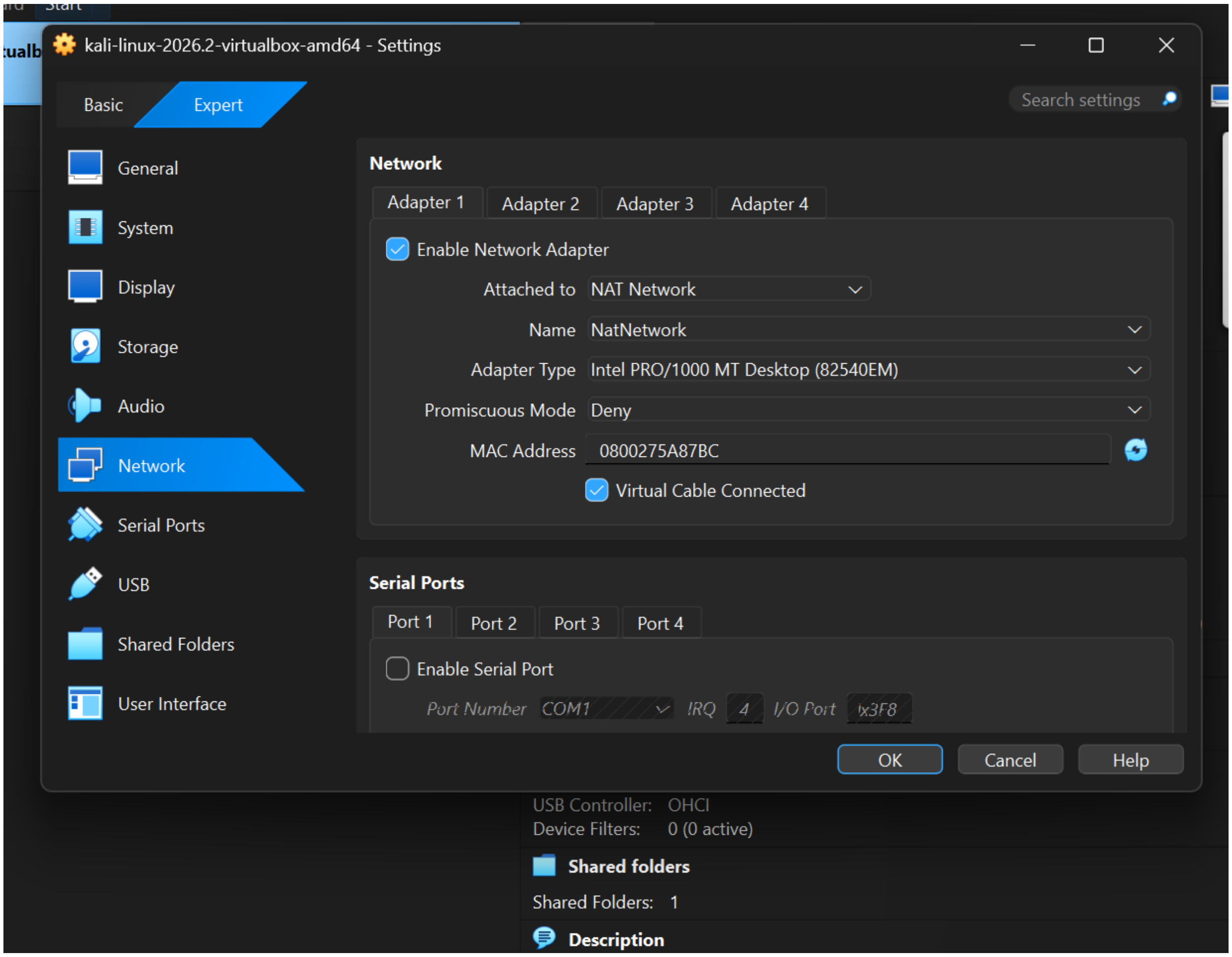


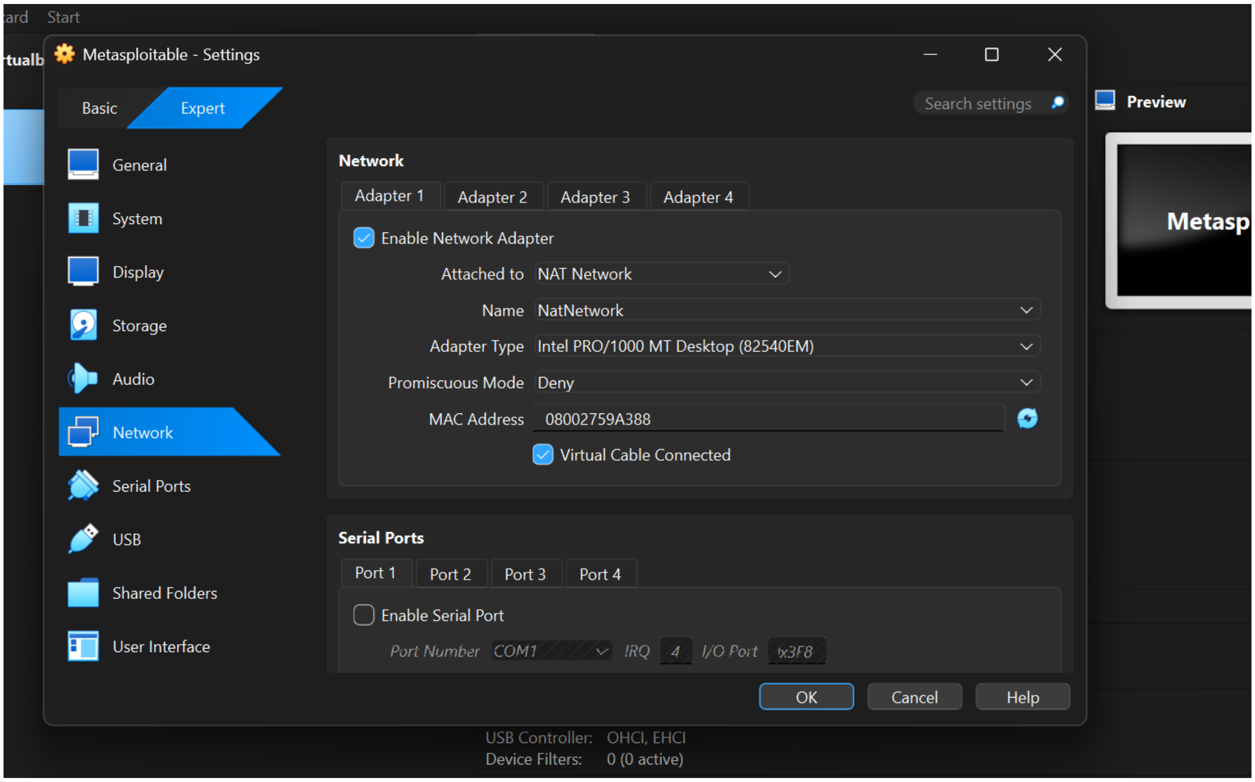
Both VMs share one network: direct communication + internet

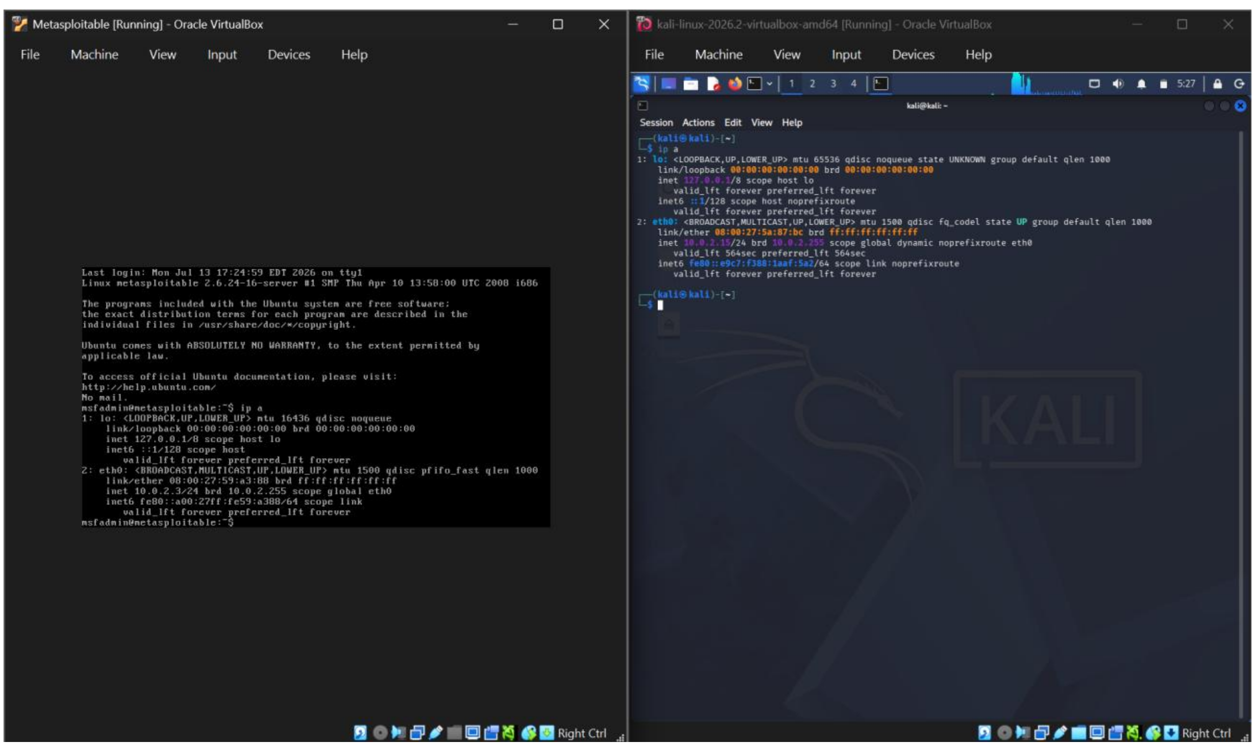
6.3: Screenshots:

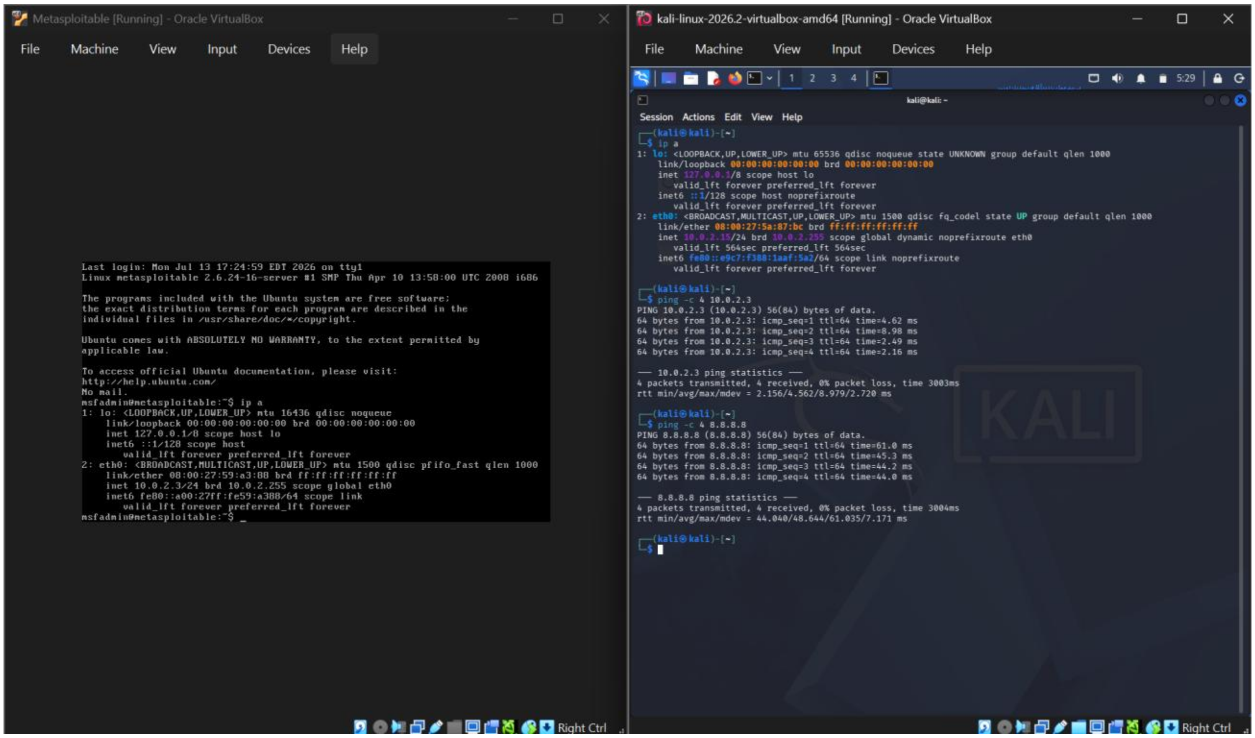


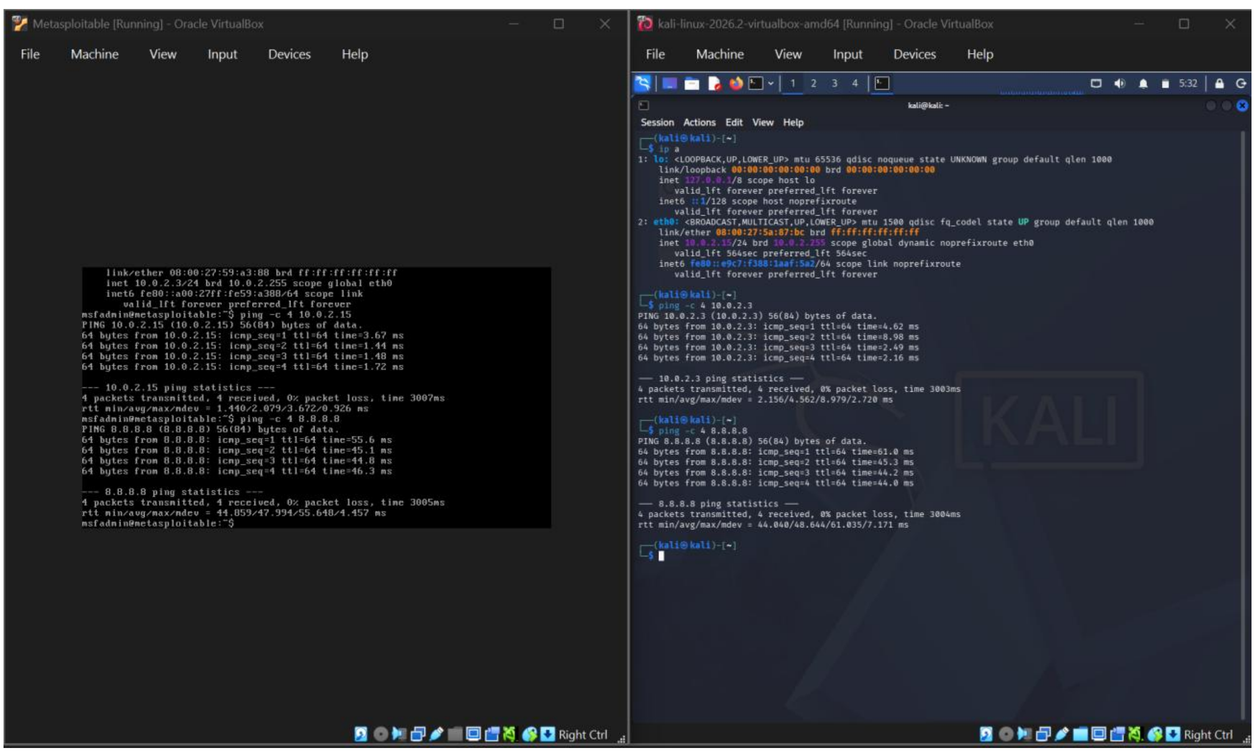



kali-linux-2026.2-virtualbox-amd64 - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to NAT Network
Name NatNetwork
Adapter Type Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode Deny
MAC Address 0800275A87BC
Virtual Cable Connected
Serial Ports
Port 1
Port 2
Port 3
Port 4
Enable Serial Port
Port Number COM1
IRQ 4
I/O Port 0x3F8
OK
Cancel
Help
USB Controller: OHCI
Device Filters: 0 (0 active)
Shared folders
Shared Folders: 1
Description


Metasploitable - Settings
Basic
Expert
Search settings
Preview
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to NAT Network
Name NatNetwork
Adapter Type Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode Deny
MAC Address 08002759A388
Virtual Cable Connected
Serial Ports
Port 1
Port 2
Port 3
Port 4
Enable Serial Port
Port Number COM1
IRQ 4
I/O Port 0x3F8
OK
Cancel
Help
USB Controller: OHCI, EHCI
Device Filters: 0 (0 active)


Metasploitable [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help





6.4: Result Table:

Test	Result
Kali IP	10.0.2.15/24
Metasploitable2 IP	10.0.2.3/24 (different address / same subnet)
Kali → Metasploitable2	Success, 2-9ms
Metasploitable2 → Kali	Success, 1-4ms
Kali → 8.8.8.8	Success, 0% loss
Metasploitable2 → 8.8.8.8	Success, 0% loss
Host → VM	Not tested

6.5: Analysis

Unlike NAT, which places each VM on its own isolated private network, NAT Network creates a single shared virtual network that multiple VMs can join simultaneously, backed by one shared DHCP server. This is why Kali and Metasploitable2 received different addresses on the same subnet (10.0.2.15 and 10.0.2.3 respectively) rather than an identical address as seen under NAT, both VMs were drawing from the same address pool on the same network.

Because both VMs now share a genuine network segment, direct communication between them succeeded in both directions, with round-trip times in the 1-9ms range. This latency, while still low, is measurably higher than the sub-millisecond self-response observed under NAT, supporting the conclusion that this traffic was genuinely traversing a network path between two separate machines rather than looping back to the same host.

Internet access also succeeded independently on both VMs, since NAT Network retains NAT's translation behavior for outbound traffic while additionally enabling inter-VM visibility, combining the benefits of both isolation from the outside network and communication within it.

6.6: Conclusion:

NAT Network resolves the core limitation of standard NAT mode by placing multiple VMs on a single shared network with a common DHCP pool, enabling genuine VM-to-VM communication while preserving independent internet access for each machine. This makes it a suitable choice for lab environments requiring multiple machines to interact directly, without sacrificing the ability to reach external resources such as software repositories or update servers.

7. Mode 3: Bridged Adapter

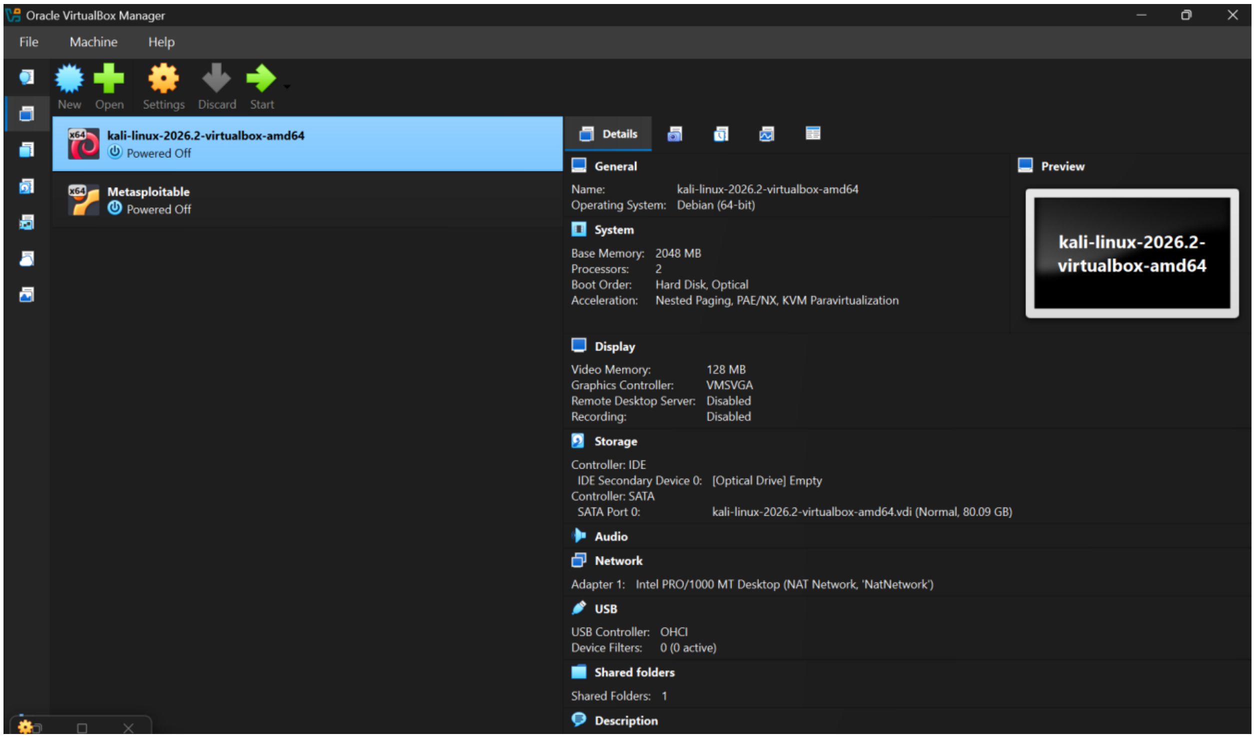
7.1: Configuration:

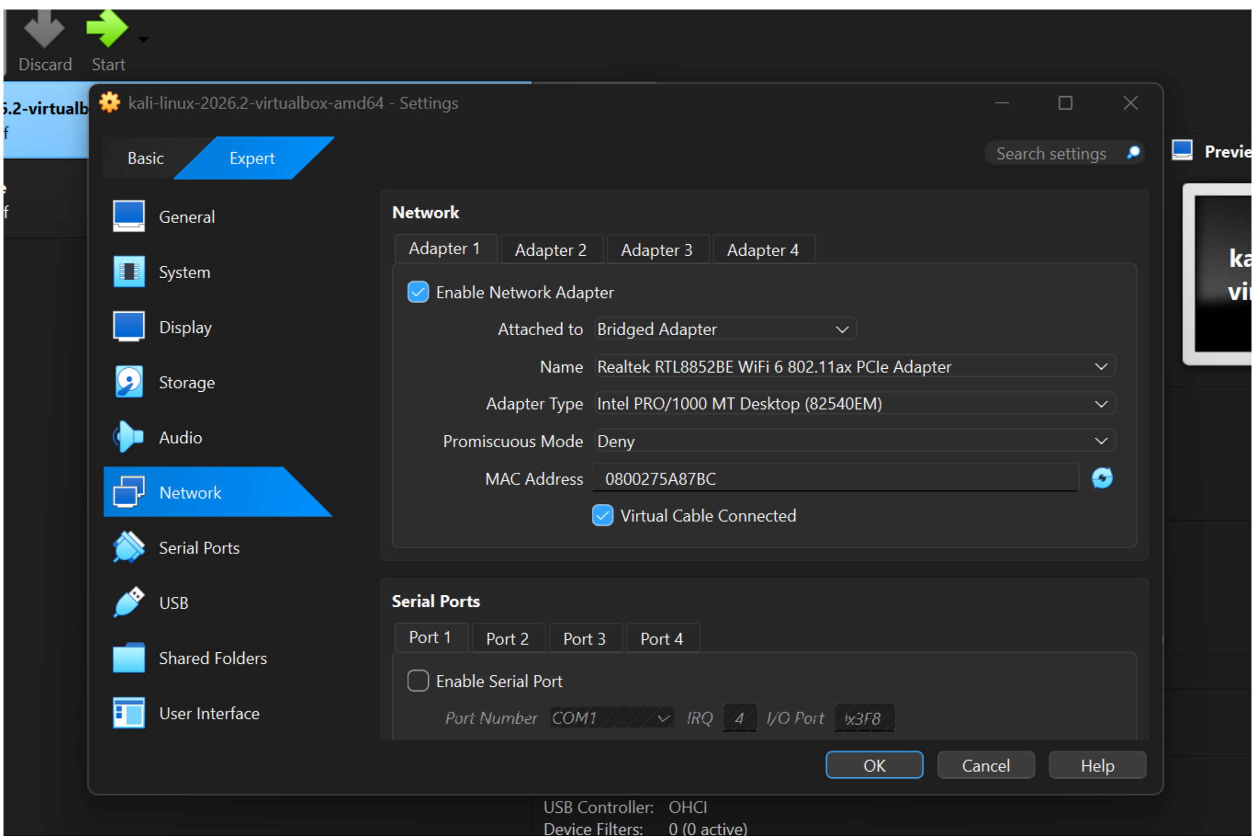
Both Kali and Metasploitable2 set to Bridged Adapter

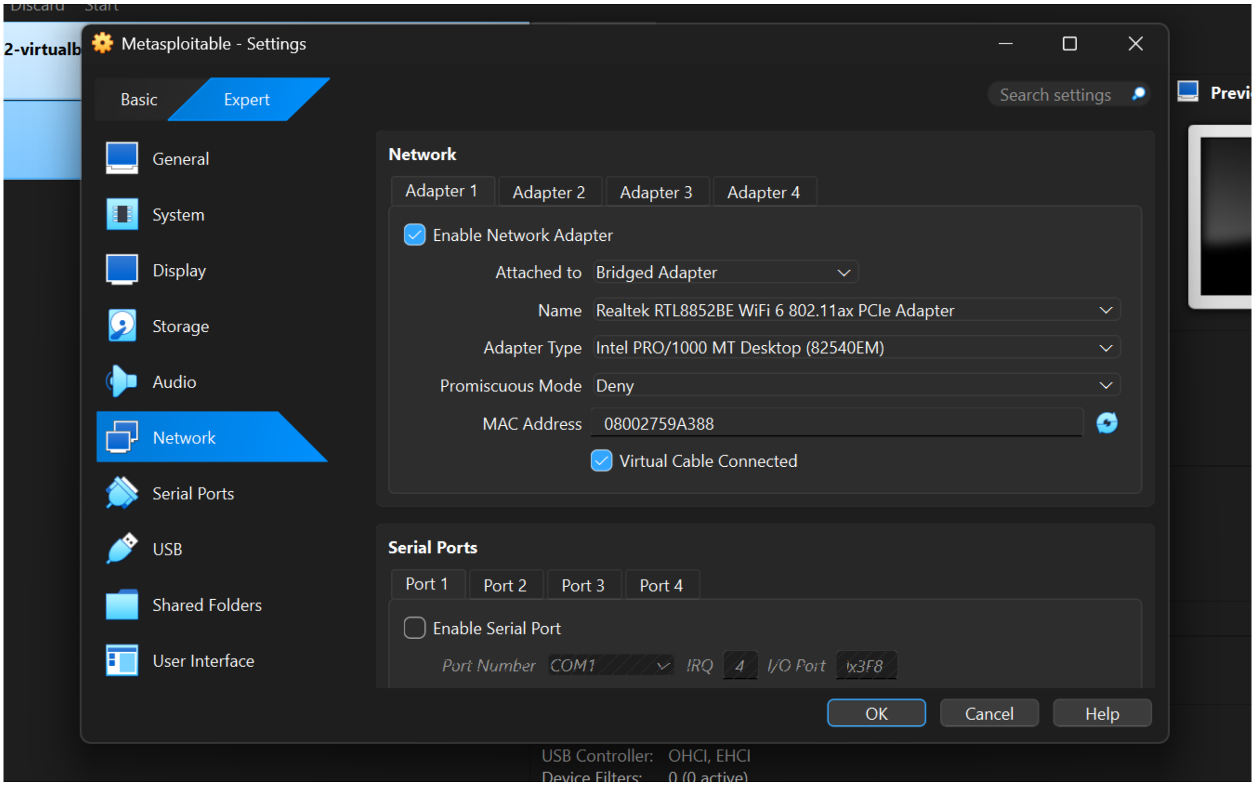
7.2: Architecture Diagram:

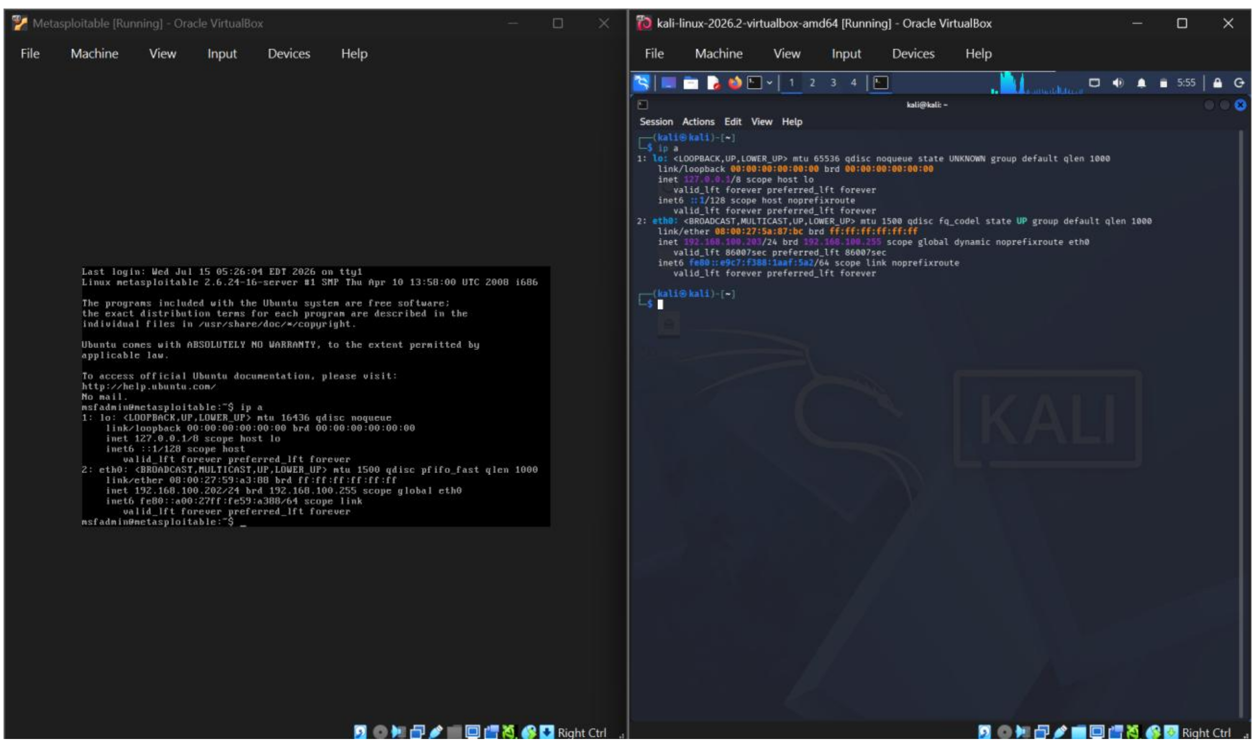
```
Physical Router /Internet
 |
Physical LAN (192.168.100.0/24)
 |
 ├── Host (192.168.100.x)
 |
 ├── Kali (192.168.100.203)
 |
 └── Metasploitable2 (192.168.100.202)

All devices are peers on the real network
```

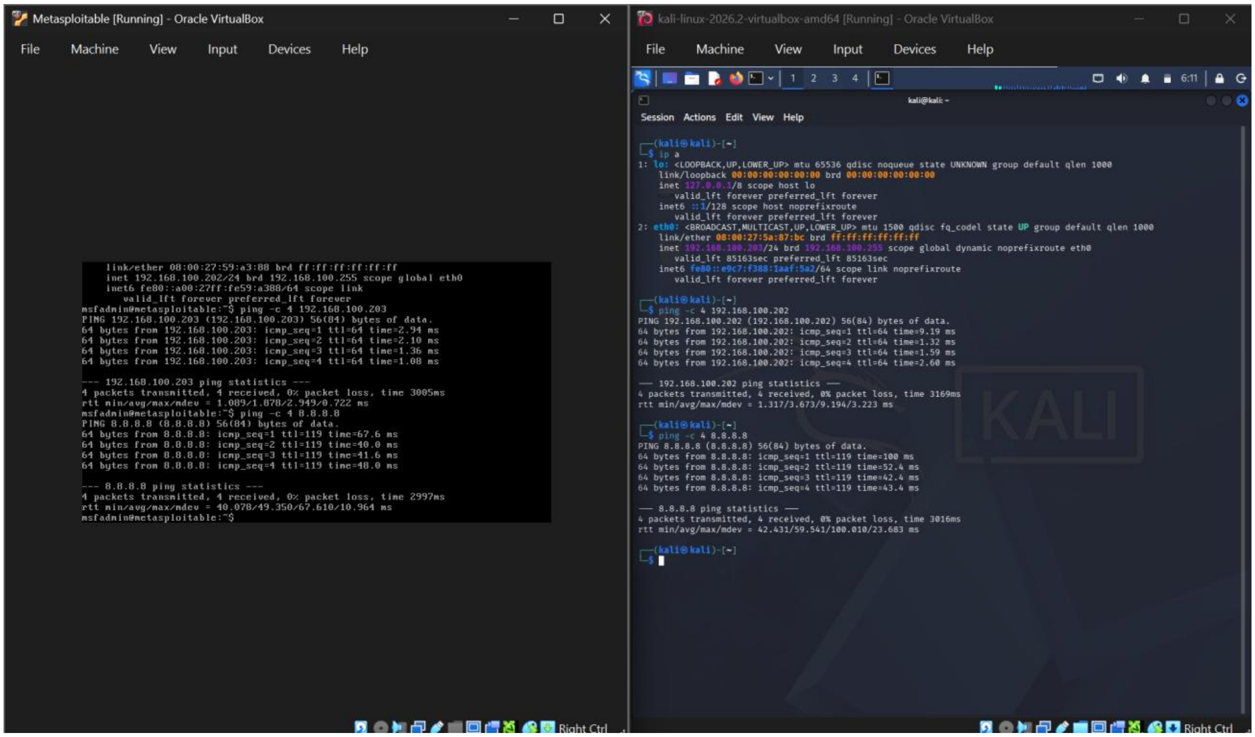
7.3: Screenshots:






Metasploitable - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to Bridged Adapter
Name Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
Adapter Type Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode Deny
MAC Address 08002759A388
Virtual Cable Connected
Serial Ports
Port 1
Port 2
Port 3
Port 4
Enable Serial Port
Port Number COM1
IRQ 4
I/O Port
OK
Cancel
Help
USB Controller: OHCI, EHCI


Metasploitable [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
Right Ctrl



7.4: Result Table:

Test	Result
Kali IP	192.168.100.203/24 (real LAN address)
Metasploitable2 IP	192.168.100.202/24 (real LAN address)
Kali → Metasploitable2	Success, 1.32-9.19ms
Metasploitable2 → Kali	Success, 1.08-2.94ms
Kali → 8.8.8.8	Success, 40-100ms, TTL=119
Metasploitable2 → 8.8.8.8	Success, 40-67ms, TTL=119
Host → VM	Not formally tested (VMs directly reachable as LAN peers by design)

7.5: Analysis

Bridged Adapter mode connects a VM's virtual network interface directly to the host machine's physical network adapter, causing the VM to appear as an ordinary device on the local network rather than existing behind VirtualBox's virtual NAT router. This behaviour was reflected in the addressing observed: both Kali and Metasploitable2 received IP addresses from the physical network's DHCP server (192.168.100.203 and 192.168.100.202), within the same address range as other devices on the LAN, rather than from a VirtualBox-managed private subnet such as 10.0.2.x.

Because both VMs were genuine peers on the same physical network, direct communication between them succeeded in both directions. Internet access also succeeded independently on both VMs through the physical network rather than through VirtualBox's NAT engine, demonstrating that each VM functioned as an independent network host.

This configuration has important security implications. Since each VM is directly connected to the physical network, it becomes visible and potentially reachable by other devices on that network. When intentionally vulnerable systems such as Metasploitable2 are used, this increases the risk of unintended exposure outside the intended lab environment.

7.6: Conclusion:

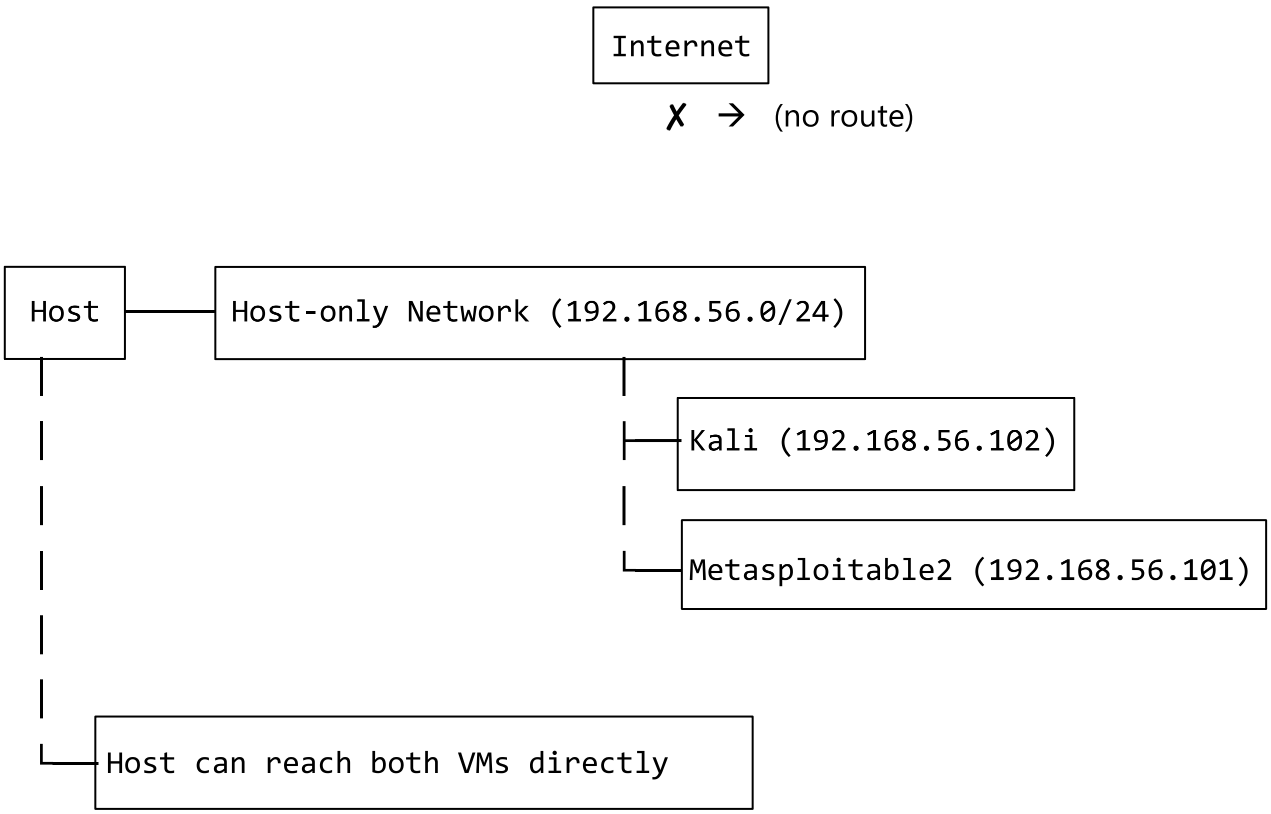
Bridged Adapter mode enables full, unrestricted communication between VMs and the wider physical network, including genuine VM-to-VM reachability and independent internet access. However, this comes at the cost of exposing the VM directly to the physical network it is bridged onto. For lab environments involving intentionally vulnerable machines, this mode should be used cautiously and only for the duration required, given the risk of exposing such machines beyond the intended lab boundary.

8. Mode 4: Host-Only Adapter

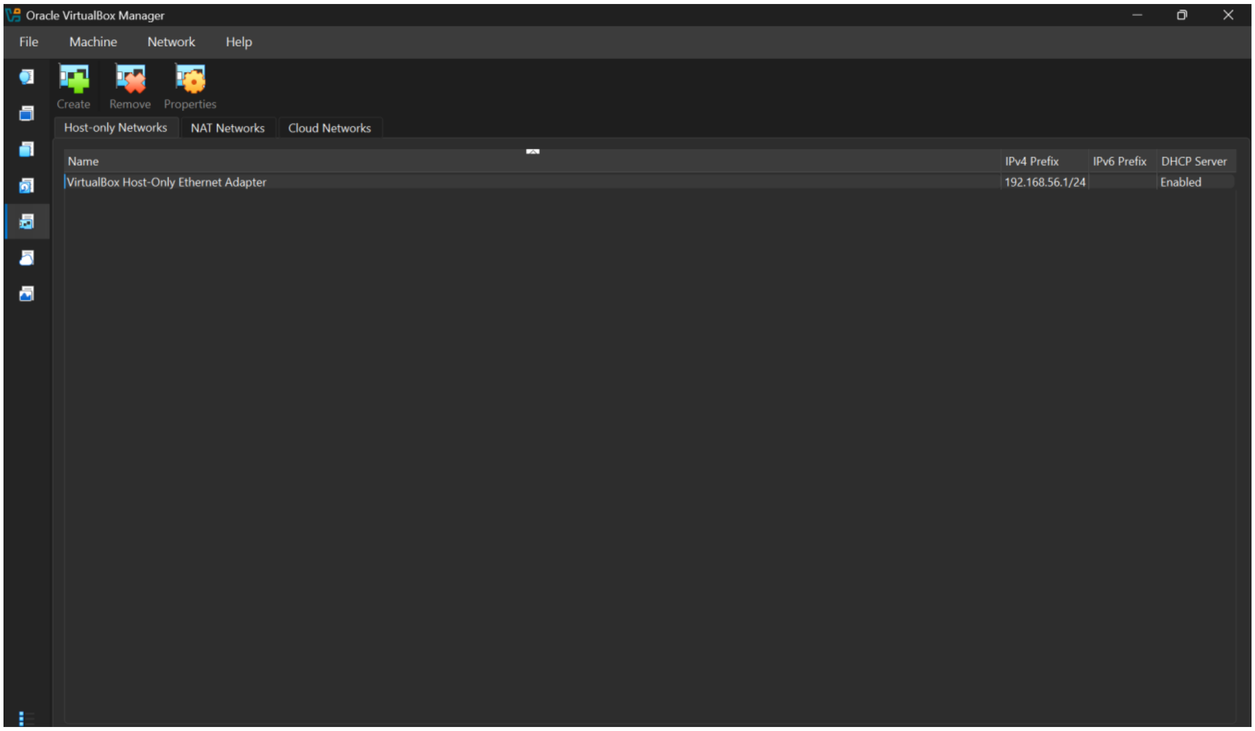
8.1: Configuration:

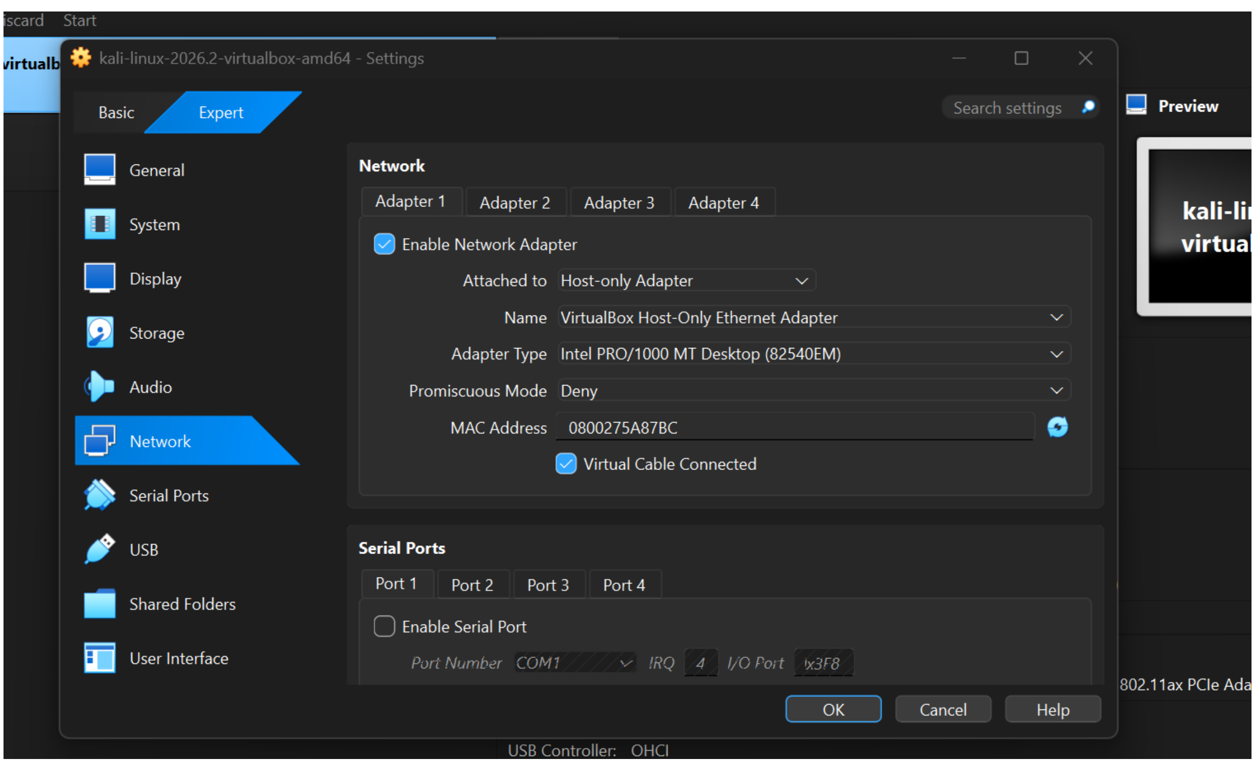
Both Kali and Metasploitable2 set to Host-Only Adapter

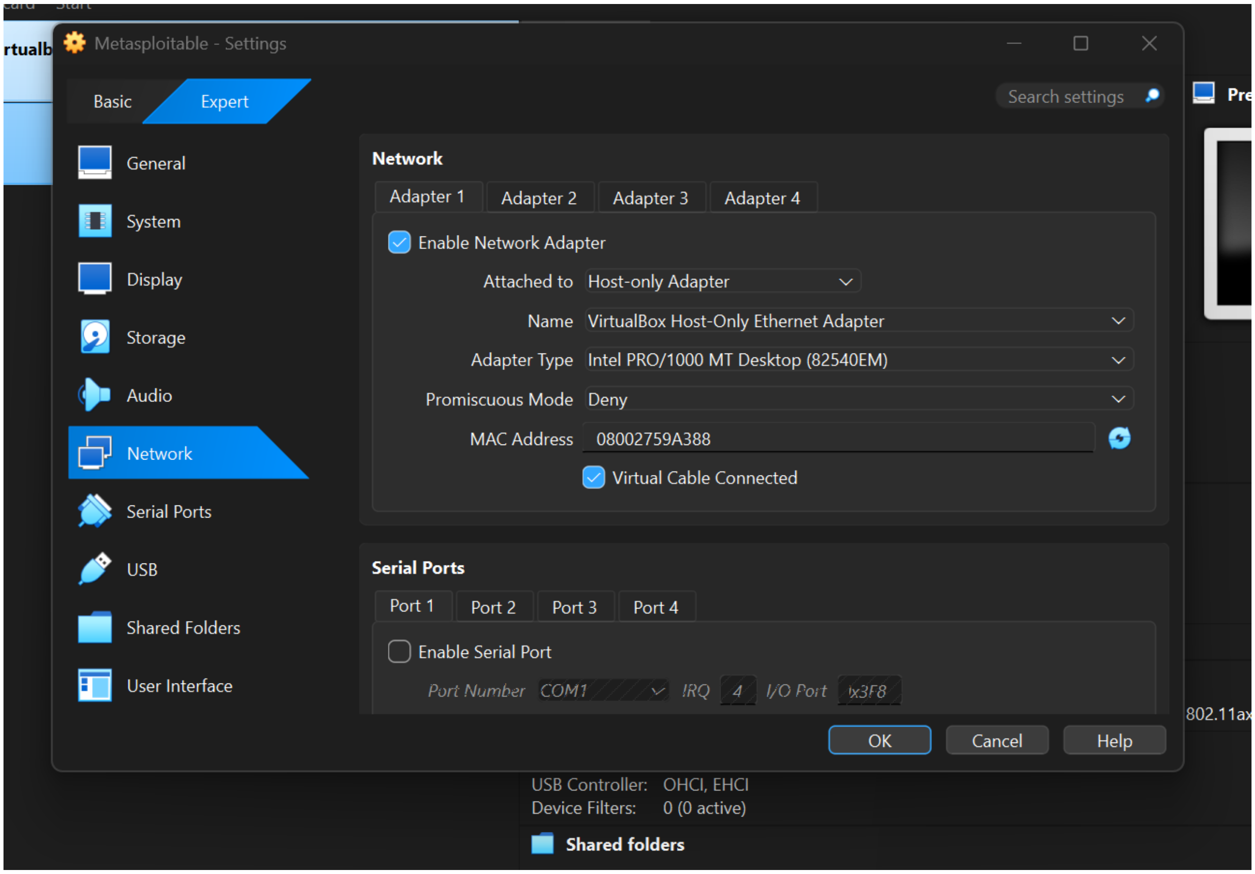
8.2: Architecture Diagram:

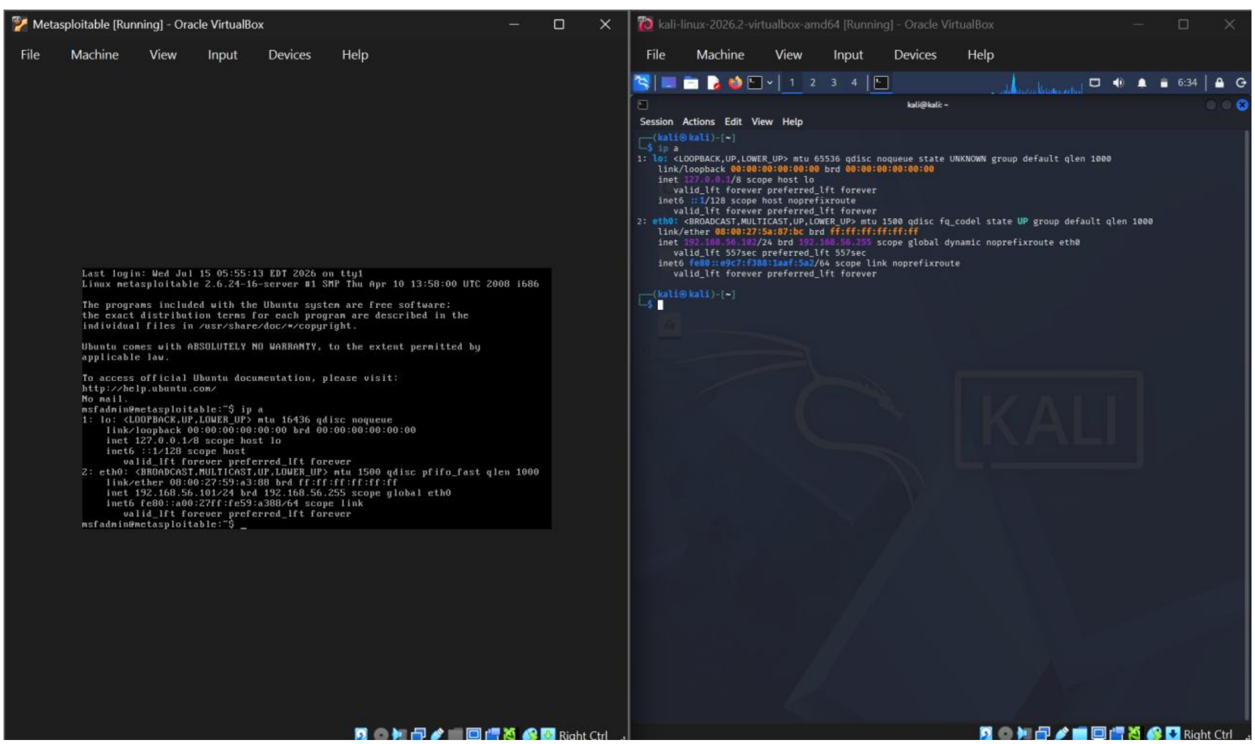


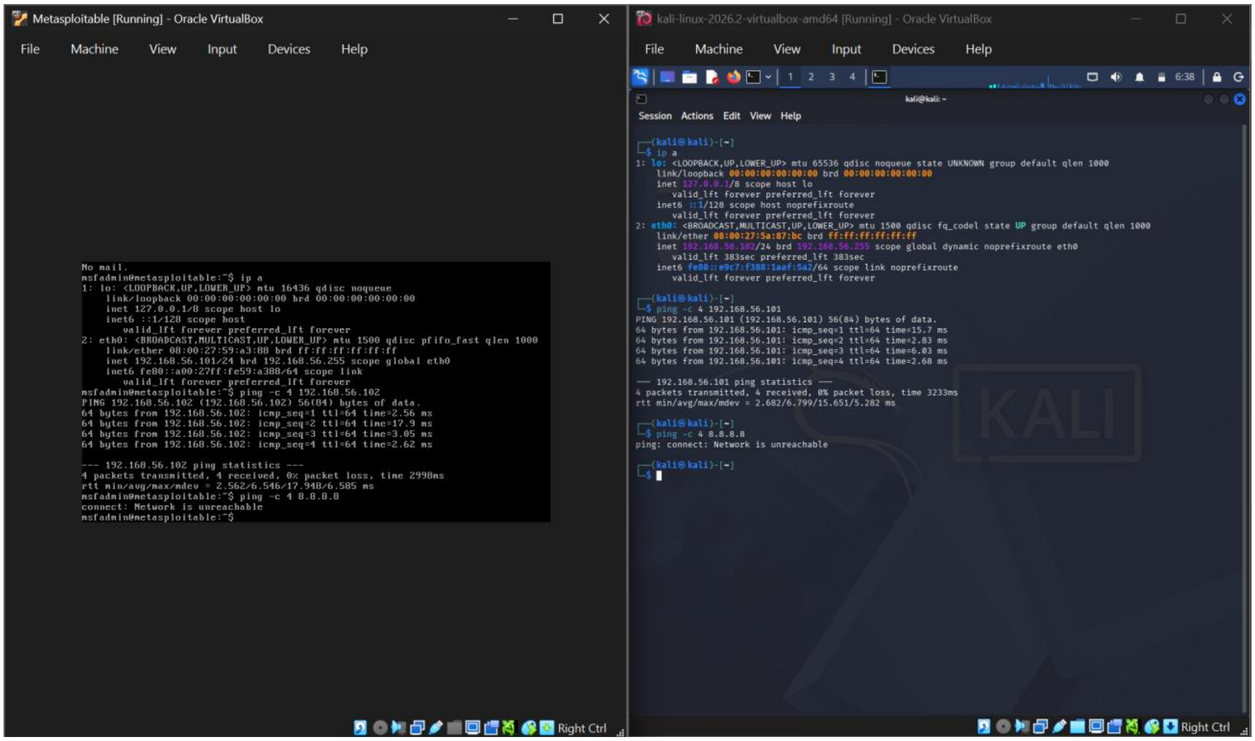
8.3: Screenshots:






Metasploitable - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to Host-only Adapter
Name VirtualBox Host-Only Ethernet Adapter
Adapter Type Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode Deny
MAC Address 08002759A388
Virtual Cable Connected
Serial Ports
Port 1
Port 2
Port 3
Port 4
Enable Serial Port
Port Number COM1
IRQ 4
I/O Port 0x3F8
OK
Cancel
Help
USB Controller: OHCI, EHCI
Device Filters: 0 (0 active)
Shared folders


Metasploitable [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
Right Ctrl



8.4: Result Table:

Test	Result
Kali IP	192.168.56.102/24 (auto-assigned via DHCP)
Metasploitable2 IP	192.168.56.101/24 (auto-assigned via DHCP)
Kali → Metasploitable2	Success, 2.68-15.7ms
Metasploitable2 → Kali	Success, 2.56-17.9ms
Kali → 8.8.8.8	Failed, "Network is unreachable"
Metasploitable2 → 8.8.8.8	Failed, "Network is unreachable"
Host → Kali	Success, avg 2ms
Windows Host → Metasploitable2	Success, avg 1ms

8.5: Analysis

Host-only Adapter mode creates an isolated virtual network shared only between the host machine and any VMs attached to it, with no bridge to the physical network and no path to the internet. Unlike Internal Network, this mode includes a functioning DHCP server, which is why both Kali and Metasploitable2 received addresses automatically (192.168.56.102 and 192.168.56.101) without requiring manual configuration.

Direct communication between the two VMs succeeded in both directions, confirming that Host-only provides genuine inter-VM connectivity in the same manner as NAT Network and Bridged.

However, attempts to reach the internet from either VM failed immediately with a "Network is unreachable" error, rather than timing out. This distinction is significant: a timeout would suggest a route existed but received no response, whereas "Network is unreachable" indicates no route to the destination exists at all, confirming that Host-only mode provides no outbound path beyond the isolated network itself.

The defining characteristic of this mode was confirmed by testing connectivity from the Windows host machine directly, which successfully reached both VMs. This capability is unique to Host-only mode among the isolated network types tested and is the practical difference between "Host-only" and "Internal Network," despite both modes lacking internet access.

8.6: Conclusion:

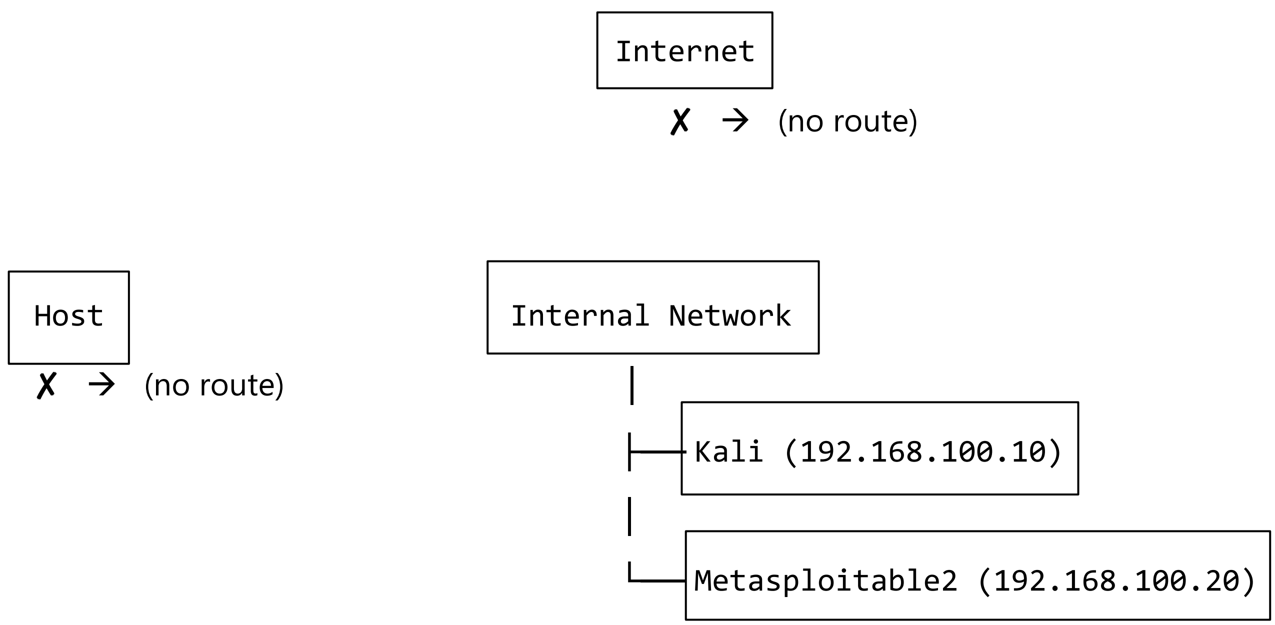
Host-only Adapter mode provides an isolated lab network with functioning inter-VM communication, automatic IP assignment via DHCP, and the added ability for the host machine to directly reach and manage the VMs on that network, without exposing them to the internet or the physical network. This makes it well suited for lab environments where the host machine itself needs visibility into the lab network, such as running host-based monitoring or management tools against the VMs.

9. Mode 5: Internal Network

9.1: Configuration:

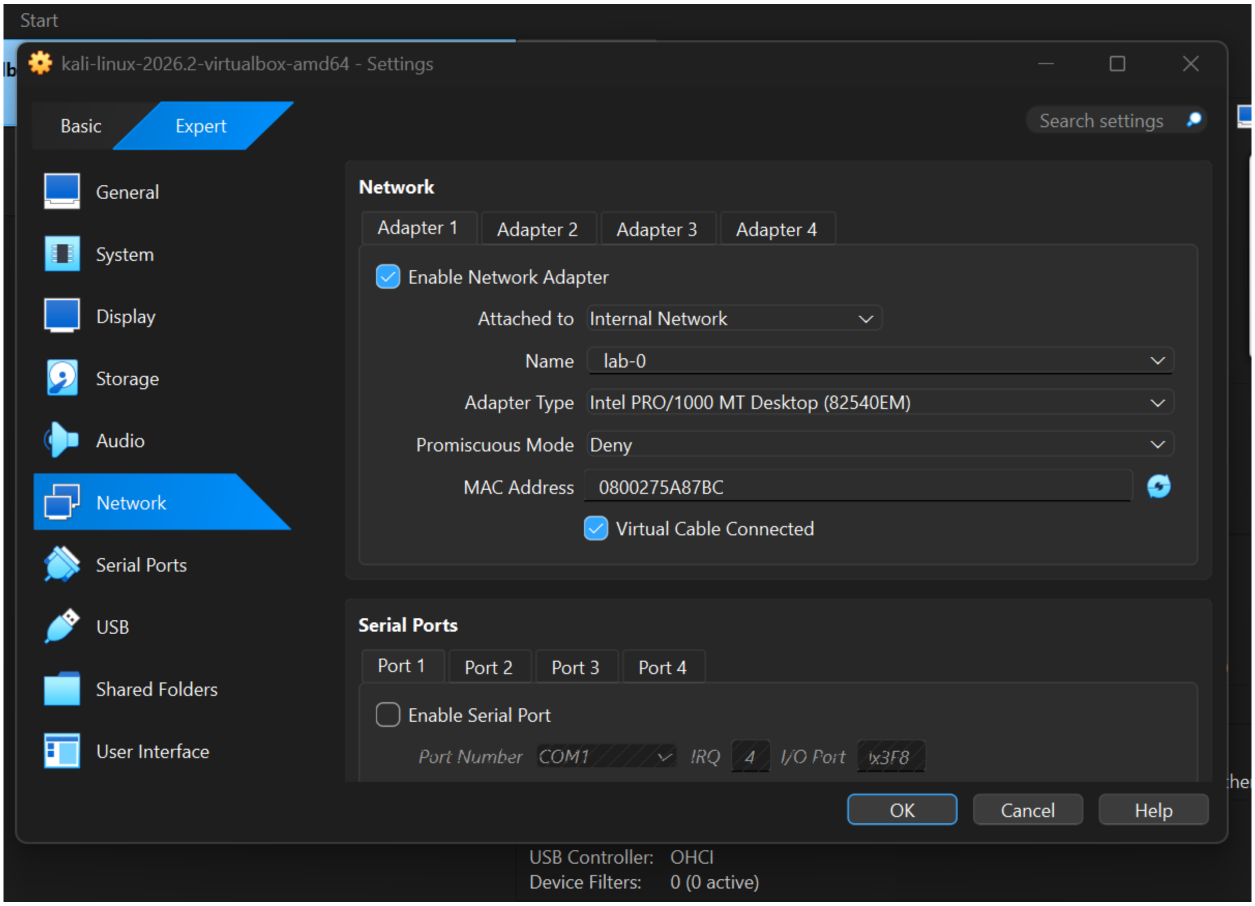
Both Kali and Metasploitable2 set to Internal Network

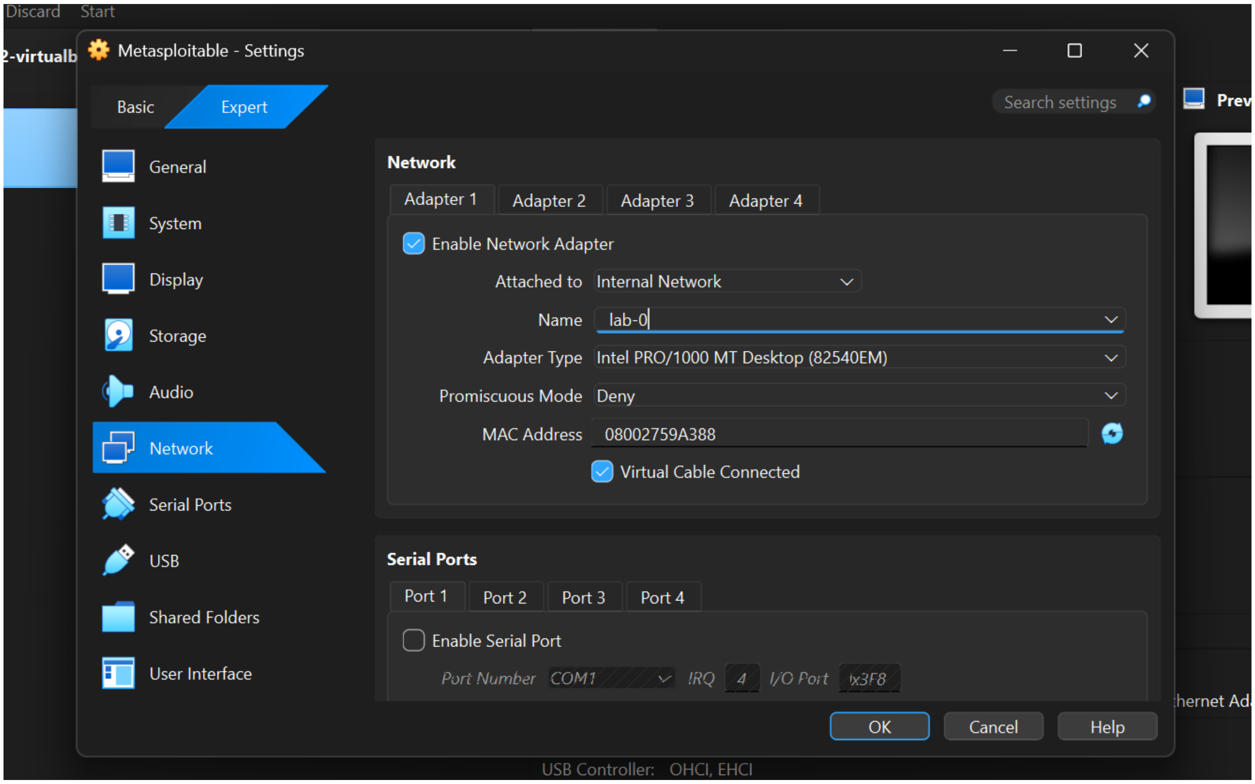
9.2: Architecture Diagram:

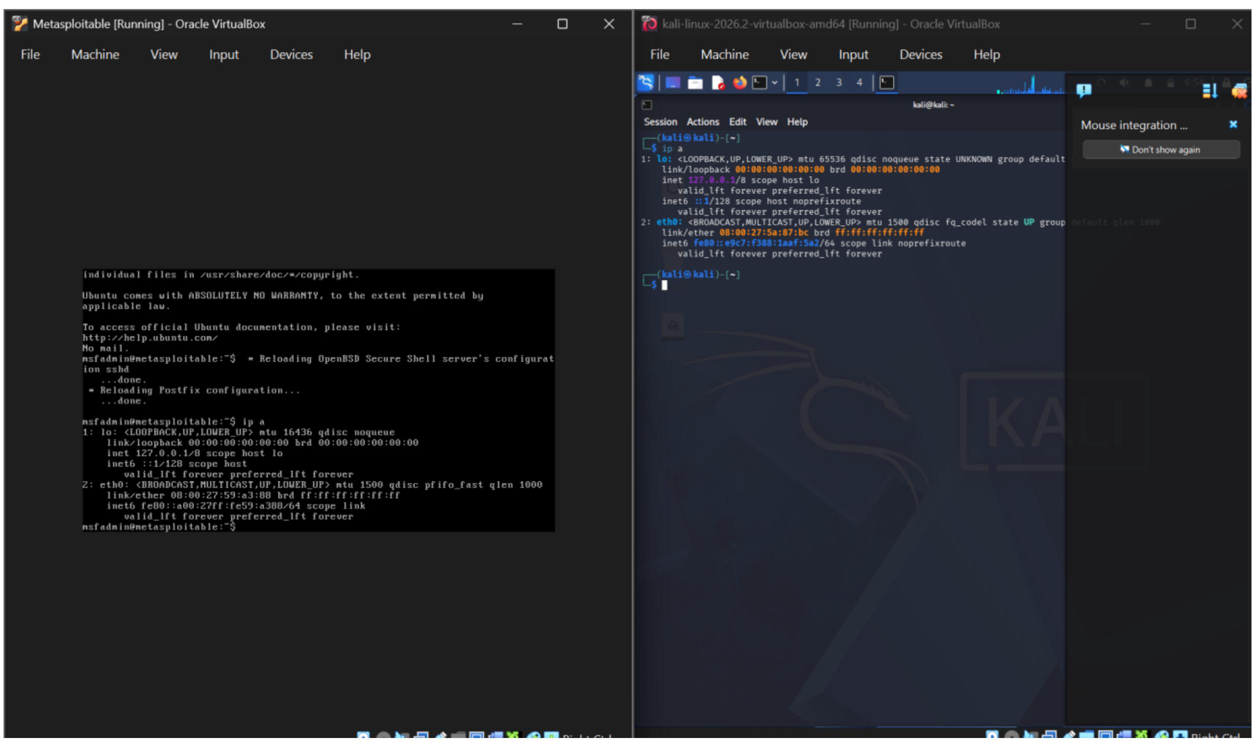


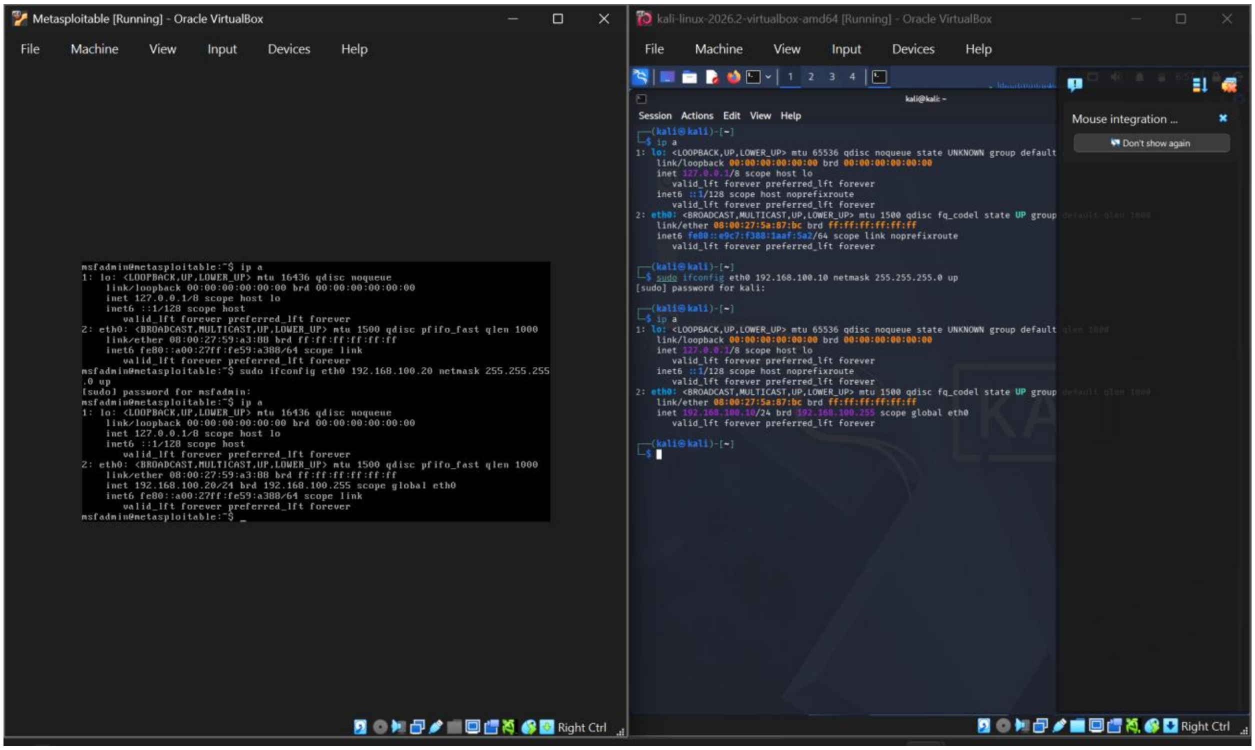
`Host is excluded entirely VM-to-VM only`

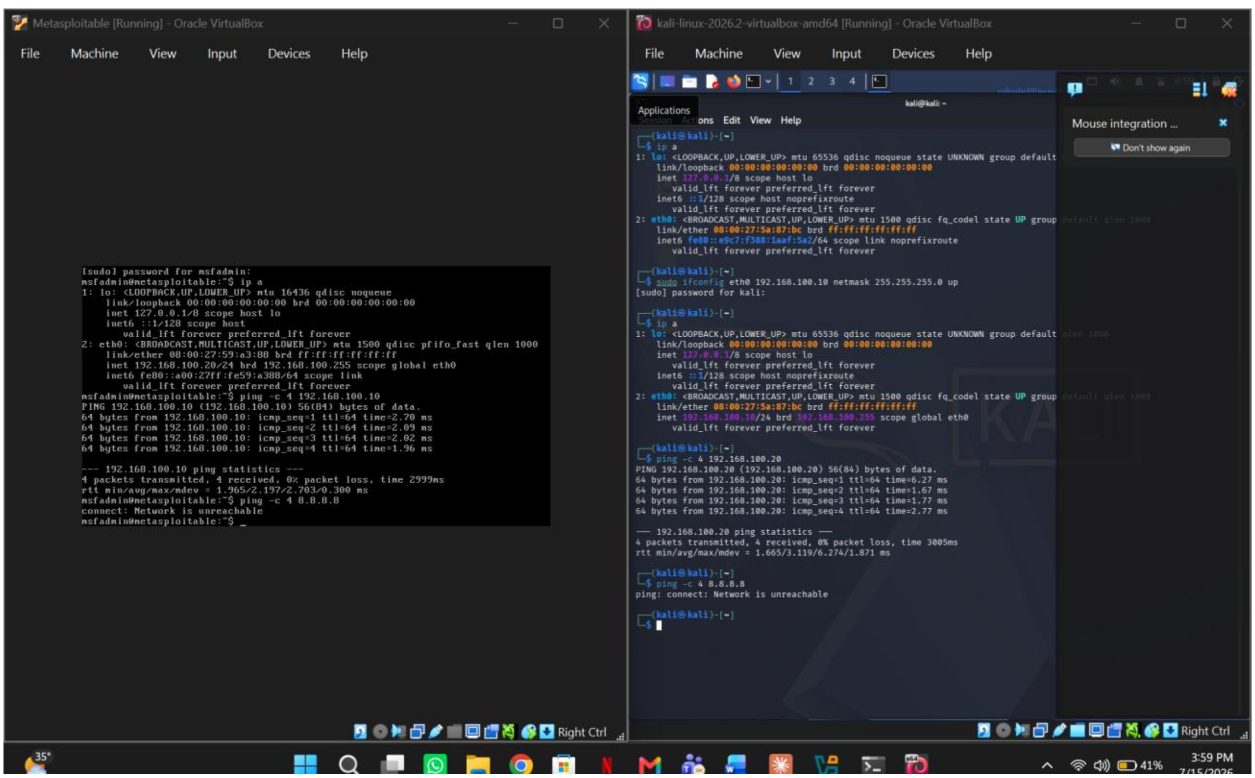
9.3: Screenshots:

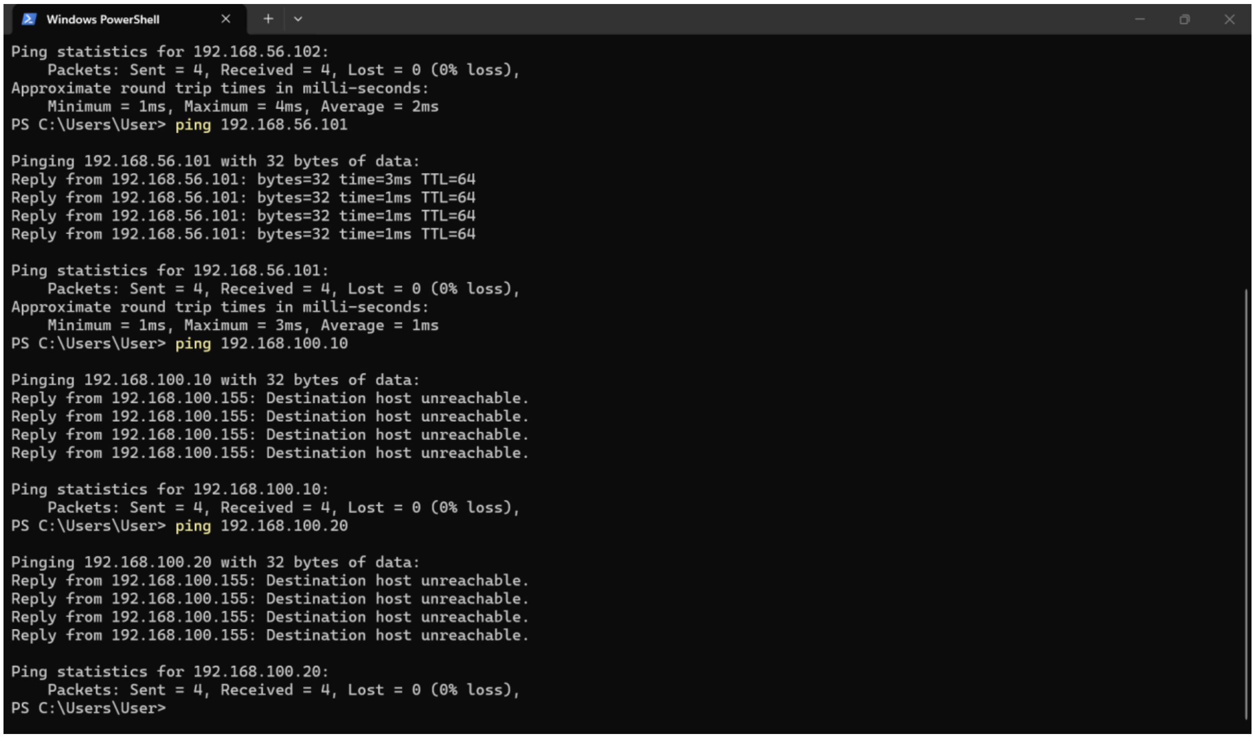



Metasploitable - Settings
Basic
Expert
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to Internal Network
Name lab-0
Adapter Type Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode Deny
MAC Address 08002759A388
Virtual Cable Connected
Serial Ports
Port 1
Port 2
Port 3
Port 4
Enable Serial Port
OK
Cancel
Help
USB Controller: OHCI, EHCI


Metasploitable [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Mouse integration ...







```
Windows PowerShell
Ping statistics for 192.168.56.102:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 4ms, Average = 2ms
PS C:\Users\User> ping 192.168.56.101

Pinging 192.168.56.101 with 32 bytes of data:
Reply from 192.168.56.101: bytes=32 time=3ms TTL=64
Reply from 192.168.56.101: bytes=32 time=1ms TTL=64
Reply from 192.168.56.101: bytes=32 time=1ms TTL=64
Reply from 192.168.56.101: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.56.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 1ms
PS C:\Users\User> ping 192.168.100.10

Pinging 192.168.100.10 with 32 bytes of data:
Reply from 192.168.100.155: Destination host unreachable.
Reply from 192.168.100.155: Destination host unreachable.
Reply from 192.168.100.155: Destination host unreachable.
Reply from 192.168.100.155: Destination host unreachable.

Ping statistics for 192.168.100.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
PS C:\Users\User> ping 192.168.100.20

Pinging 192.168.100.20 with 32 bytes of data:
Reply from 192.168.100.155: Destination host unreachable.
Reply from 192.168.100.155: Destination host unreachable.
Reply from 192.168.100.155: Destination host unreachable.
Reply from 192.168.100.155: Destination host unreachable.

Ping statistics for 192.168.100.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
PS C:\Users\User>
```

9.4: Result Table:

Test	Result
Kali IP	No automatic address; manually assigned 192.168.100.10/24
Metasploitable2 IP	No automatic address; manually assigned 192.168.100.20/24
Kali → Metasploitable2	Success (after manual IP assignment)
Metasploitable2 → Kali	Success (after manual IP assignment)
Kali → 8.8.8.8	Failed, "Network is unreachable"
Metasploitable2 → 8.8.8.8	Failed, "Network is unreachable"
Host → Kali	Failed, "Destination host unreachable" (request redirected to host's real gateway)
Windows Host → Metasploitable2	Failed, "Destination host unreachable" (request redirected to host's real gateway)

9.5: Analysis

Internal Network mode provides the same VM-to-VM isolation as Host-only mode but excludes even the host machine from the network entirely and does not include a DHCP server. This absence of DHCP was evident when Metasploitable2 initially received no IPv4 address at all after being attached to this network, requiring static IP addresses to be assigned manually to both VMs before communication could be tested.

Once static addresses were configured, direct communication between Kali and Metasploitable2 succeeded in both directions, confirming that the virtual network itself functioned correctly once addressing is resolved manually. Internet access failed on both VMs with the same "Network is unreachable" error observed under Host-only mode, consistent with Internal Network also providing no outbound route.

The distinguishing result for this mode was the host-to-VM test. Unlike Host-only mode, attempts to ping either VM from the Windows host failed, with replies returned from the host's own default gateway rather than from the VMs themselves. This indicates that the host machine has no route to the Internal Network at all; ping requests directed at the VMs' addresses were instead redirected to the host's real network gateway, which correctly reported the destination as unreachable, since no such device exists on the host's actual network.

9.6: Conclusion:

Internal Network mode provides the strictest isolation of the five modes tested, permitting communication only between VMs attached to the same named internal network, while excluding both the host machine and the internet entirely. This makes it the most suitable mode for lab environments requiring complete isolation from any real network, at the cost of requiring manual IP configuration due to the absence of a DHCP server and losing the host-to-VM management capability available under Host-only mode.

10. Overall Comparative Summary Table

Mode	VM-to-VM	Internet	Host → VM	IP Assignment
NAT	✗ (self-loop)	✓	Not applicable	Automatic, identical per VM
NAT Network	✓	✓	Not tested	Automatic, shared pool
Bridged	✓	✓	N/A (real LAN peer)	Automatic, real LAN DHCP
Host-only	✓	✗	✓	Automatic (DHCP present)
Internal Network	✓ (manual)	✗	✗	Manual (no DHCP)

11. Overall Conclusion

Overall Conclusion

This investigation demonstrated that the choice of VirtualBox network mode is a fundamental design decision rather than a simple configuration setting. Each networking mode offers a different balance between connectivity, isolation, and accessibility, making it suitable for different scenarios. Selecting an inappropriate mode can prevent required communication between virtual machines, expose systems to unnecessary security risks, or produce unexpected network behavior.

Rather than adopting a single network mode for all future work, the results of this investigation show that the network configuration should be selected according to the specific objectives and requirements of each laboratory exercise:

- **NAT** is suitable when a single virtual machine requires internet access without communicating with other virtual machines.
- **NAT Network** is suitable when virtual machines must communicate with one another while maintaining independent internet access.
- **Host-only Adapter** is suitable when communication is required between the virtual machines and the host system, but internet access is unnecessary or intentionally disabled.
- **Bridged Adapter** is suitable only when interaction with the physical network is required and should be used cautiously, particularly when working with intentionally vulnerable systems.
- **Internal Network** is suitable when complete isolation is required, including isolation from the host machine, such as for highly contained security experiments.

Accordingly, the network mode for future laboratories will be chosen on a case-by-case basis to match each exercise's connectivity, isolation, and security requirements.

This investigation also highlights the importance of validating virtual networking behaviour through practical experimentation rather than relying solely on theoretical descriptions or documentation. Empirical testing provides a clearer understanding of how each networking mode behaves in practice and ensures that the selected configuration supports the intended objectives of the laboratory environment.